Check for updates

Concise Report

Palladium catalysis, Propargylic alcohols, Boronic acids, Allenes, Biological activities

CJC Chinese Journal of Chemistry 中国化学 · An International Journal

Palladium-Catalyzed Coupling of Propargylic Alcohols with Boronic Acids under Ambient Conditions†

Anni Qin,[a] Hui Qian,[a] Qin Chen,*[,a] and Shengming Ma*[,a,b]

[a] *Research Center for Molecular Recognition and Synthesis, Department of Chemistry, Fudan University, 220 Handan Lu, Shanghai 200433, China*
[b] *State Key Laboratory of Organometallic Chemistry, Shanghai Institute of Organic Chemistry, Chinese Academy of Sciences, 345 Lingling Lu, Shanghai 200032, China*

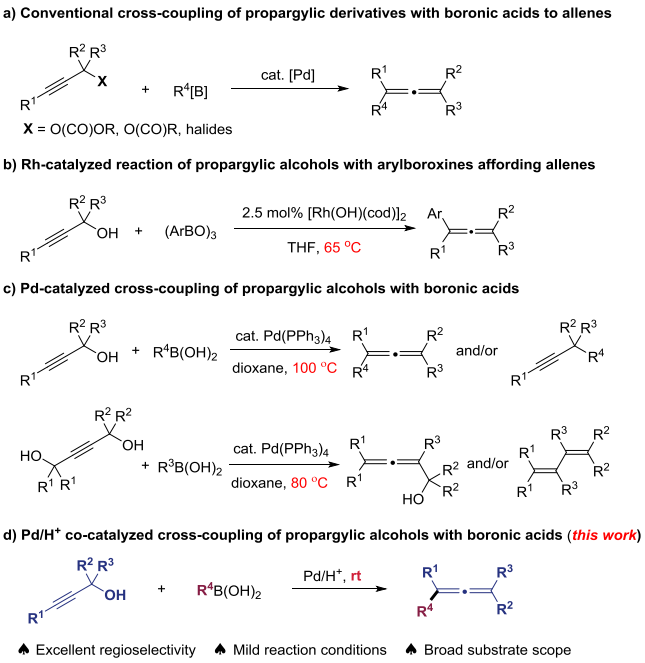
Cite this paper: *Chin. J. Chem.* **2020**, *38*, 372—382. **DOI: 10.1002/cjoc.201900442**

Summary of main observation and conclusion A highly efficient palladium-catalyzed direct coupling of propargylic alcohols with organoboronic acids to synthesize tri- and tetra-substituted allenes has been developed under mild reaction conditions. Many useful functional groups are tolerated in this process with high to excellent yields. Preliminary biological studies showed that several tri- and tetra-substituted allenes exhibited potent anti-diabetic activities.

Background and Originality Content

Allenes are important synthetic building blocks in organic synthesis and many natural products and pharmaceutical drugs contain allene motifs.[1,2] Consequently, many studies have been focused on the synthesis of different types of allenes.[1g,3] Transition metal-catalyzed cross-coupling reactions are powerful tools for C—C bond formation.[4] The Suzuki coupling reaction has been widely exploited in last few decades.[5] Pd-catalyzed coupling of propargylic compounds with organoboronic acids is one of the most straightforward approaches for allene synthesis.[6] However, reactive substrates bearing a good leaving group such as propargylic halides, esters or carbonates were required in these protocols (Scheme 1a).[6]

Scheme 1 Transition metal-catalyzed cross-coupling to synthesize allenes



Coupling of propargylic alcohols has less been reported due to the difficulty encountered in the step of oxidative addition with a transition metal catalyst.[7-9] Dou and co-workers reported Rh-catalyzed cross-coupling of propargylic alcohols with arylboroxines affording allenes via S_N2'-type mechanism (Scheme 1b).[7a] Yoshida and Ihara found the Pd-catalyzed direct coupling of propargylic alcohols with arylboronic acids affording allenes and/or alkynes could be achieved at 100 °C (Scheme 1c, up).[6j] Sherburn and co-workers developed Pd-catalyzed direct coupling of 2-butynyl 1,4-diols with boronic acids generating allenes and/or dienes at 80 °C (Scheme 1c, down).[6k] Recently, our group put forward a new concept of utilizing a catalytic amount of Brønsted acid to form a transient leaving group from the hydroxy group, and successfully achieved the atom-economic synthesis of 2,3-allenoates and 2,3-allenoic acids via the carboxylation with CO.[8] With our goal of developing more efficient methods in the synthesis of allenes with simple starting substrates, we propose a direct coupling of organoboronic acids with propargylic alcohols under mild conditions under the co-catalysis of Pd(0) and an organic acid (Scheme 1d).

Results and Discussion

We began our investigation with the propargylic alcohol **1a** which bears one phenyl and one methyl group on the α-carbon and phenylboronic acid **2a** to identify the ligands for the reaction at room temperature (Scheme 2). We initially used *o*-(diphenylphosphino) benzaldehyde (**L1**), which performed efficiently in the system of propargylic carbonates.[6g] Unfortunately, 94% of propargylic alcohol **1a** was recovered and the desired product **3aa** was not formed. Triphenylphosphine (**L2**) was then employed as the ligand and the desired allene product **3aa** was formed in only 10% yield. Tri(*o*-tolyl)phosphine (**L3**) provided 5% yield of **3aa** with 69% recovery of **1a**.[6h,6i] Zheda-phos (**L4**) also failed to give the allene product.[9] We next investigated the effect of bidentated phosphine ligand. However, both BINAP (**L5**) and DPEPhos (**L6**) could not afford allene product **3aa**. To our surprise, using of $Pd(PPh_3)_4$ instead of $Pd_2(dba)_3{\cdot}CHCl_3$ could give 91% yield allene product **3aa** without any recovery of **1a**.

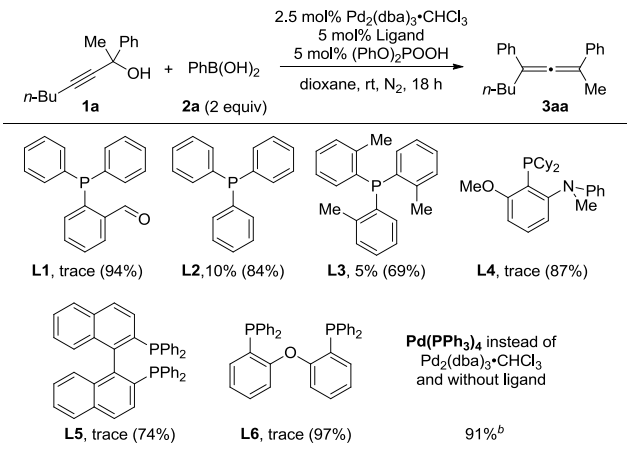
After identifying $Pd(PPh_3)_4$ as the best catalyst, other reaction parameters were then studied (Table 1). Only 10% of desired

*E-mail: chenq@fudan.edu.cn; masm@sioc.ac.cn
†Dedicated to Professor Mingyuan He on the occasion of his 80th birthday.

For submission: https://mc.manuscriptcentral.com/cjoc
For articles: https://onlinelibrary.wiley.com/journal/16147065

© 2020 SIOC, CAS, Shanghai, & WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim

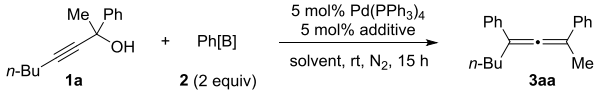
Scheme 2 Ligand screening[a]



[a] The reaction was conducted with 0.2 mmol of **1a**, 0.4 mmol of **2a**, 2.5 mol% of $Pd_2(dba)_3 \cdot CHCl_3$, 5 mol% ligand, and 5 mol% of $(PhO)_2POOH$ in 1.0 mL of anhydrous dioxane at rt for 18 h under N_2 atmosphere. The yield and recovery were determined by ^1H NMR analysis, and the recovery of **1a** is shown in the parentheses. [b] 5 mol% $Pd(PPh_3)_4$ was used and reaction time is 13 h.

allene product was observed without adding $(PhO)_2POOH$ as additive (entry 1). And no reaction occurred when 4 equiv of water were added as the additive (entry 3). Next, solvent was found to be crucial for this coupling process. Moderate efficiency was observed in THF, DME, and toluene (entries 4—6), while no reaction occurred in CH_3CN (entry 7). The yield dropped to 60% when reducing the amount of $PhB(OH)_2$ to 1.5 equiv (entry 8). We also investigated the reactivity of other boron reagents (entries 9—11): PhBpin, $(PhBO)_3$, and $PhBF_3K$ failed to afford the desired product. Thus, optimal conditions have been defined as 5 mol% of $Pd(PPh_3)_4$ and 5 mol% of $(PhO)_2POOH$ in dioxane at room temperature under N_2 atmosphere.

Table 1 Optimization of the reaction conditions[a]

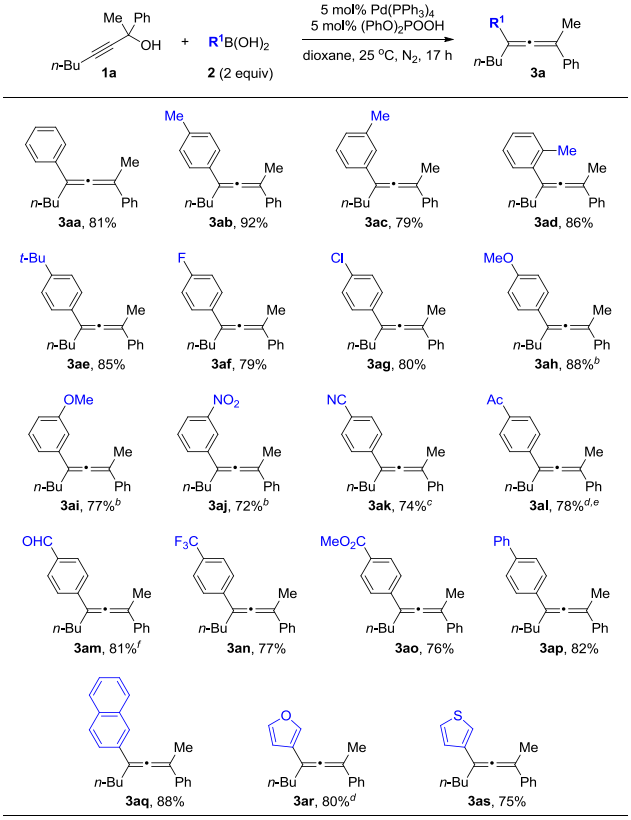


Entry	Solvent	Additive	Ph[B]	NMR yield of **3aa**/%	Recovery of **1a**/%
1[b]	dioxane	none	$PhB(OH)_2$	10	90
2[b]	dioxane	$(PhO)_2POOH$	$PhB(OH)_2$	91	trace
3[b]	dioxane	4 equiv H_2O	$PhB(OH)_2$	trace	100
4	THF	$(PhO)_2$ POOH	$PhB(OH)_2$	44	38
5	DME	$(PhO)_2$ POOH	$PhB(OH)_2$	38	38
6	toluene	$(PhO)_2$ POOH	$PhB(OH)_2$	44	24
7	MeCN	$(PhO)_2$ POOH	$PhB(OH)_2$	trace	85
8[c]	dioxane	$(PhO)_2$ POOH	$PhB(OH)_2$	60	16
9[d]	dioxane	$(PhO)_2$ POOH	PhBpin	trace	98
10[d]	dioxane	$(PhO)_2$ POOH	$(PhBO)_3$	trace	91
11[d]	dioxane	$(PhO)_2$ POOH	$PhBF_3K$	trace	86

[a] The reaction was conducted with 0.2 mmol of **1a**, 0.4 mmol of **2**, 5 mol% of $Pd(PPh_3)_4$, and 5 mol% of additive in 1.0 mL of anhydrous solvent at rt for 15 h under N_2 atmosphere. The yield and recovery were determined by ^1H NMR analysis. [b] Reaction time is 13 h. [c] 1.5 equiv $PhB(OH)_2$ was used. [d] Reaction time is 18.5 h.

With the optimized condition in hand, we examined the scope of boronic acids with propargylic alcohol **1a**. As shown in Scheme 3, a wide range of aryl boronic acids with different substituents at different positions all reacted smoothly with **1a** to form the corresponding tetrasubstituted allene products **3** with good to excellent efficiency. The reaction is amenable to both electron-withdrawing and electron-donating groups. Aryl boronic acid bearing *ortho*-, *meta*-, or *para*-methyl substitutions gave the products in high yields under optimal reaction conditions (**3ab**—**3ad**). Other electron-donating substituents such as *tert*-butyl and methoxy also afforded allene products in good yields (**3ae**, **3ah**, and **3ai**). The halo-substituted (F, Cl) aryl boronic acid also worked (**3af** and **3ag**). The electron-withdrawing groups such as nitro, cyano, acetyl, formyl in aryl were all tolerated under ambient conditions (**3aj**—**3am**). The useful functional groups such as trifluoromethyl and ester afforded the corresponding allenes successfully (**3an** and **3ao**). 4-Biphenylboronic acid and 2-naphthyl boronic acid were also suitable (**3ap** and **3aq**). Heterocycles such as furan and thiophene may also be tolerated with 80% and 75% yields, respectively (**3ar** and **3as**).

Scheme 3 Scope of boronic acids with **1a**[a]



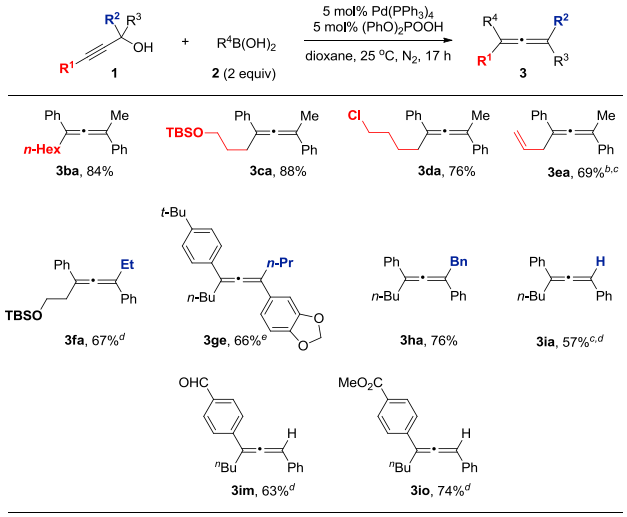
[a] The reaction was conducted with 1.0 mmol of **1a**, 2.0 mmol of **2**, 5 mol% of $Pd(PPh_3)_4$, and 5 mol% of $(PhO)_2POOH$ in 5.0 mL of anhydrous dioxane at 25 °C for 17 h under N_2 atmosphere. [b] Reaction time is 36 h. [c] The reaction was conducted at 40 °C. [d] 10 mol% of $(PhO)_2POOH$ was used. [e] Reaction time is 24 h. [f] Reaction time is 19 h.

As shown in Scheme 4, a wide range of propargylic alcohols may participate in the efficient process to form the desired allenes. Propargylic alcohols with different chain lengths of alkyl groups did not affect the yield under standard conditions (**3ba**). Alkyl chain with different functional groups, such as silyl-protected alcohol (**3ca** and **3fa**), halide (**3da**) and vinyl (**3ea**) were well tolerated in this transformation. Ethyl, propyl and benzyl substitutions on the α-carbon of propargylic alcohol performed well to afford corresponding allenes (**3fa**, **3ge**, and **3ha**). It is worth noting that secondary propargylic alcohol **1i** also worked with different arylboronic acids to afford trisubstituted allenes with moderate to good yields (**3ia**, **3im** and **3io**).

 © 2020 SIOC, CAS, Shanghai, & WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim

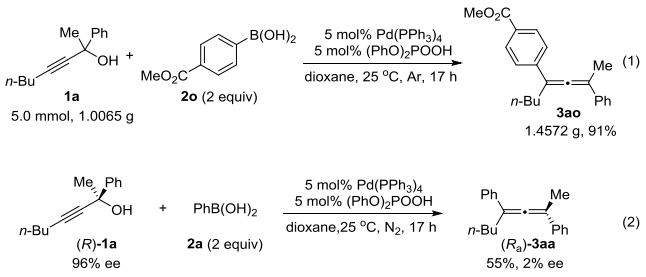
16147065, 2020, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cjoc.201900442 by Zhejiang University, Wiley Online Library on [01/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Scheme 4 Substrate scope[a]



[a] The reaction was conducted with 1.0 mmol of **1a**, 2.0 mmol of **2**, 5 mol% of $Pd(PPh_3)_4$, and 5 mol% of $(PhO)_2POOH$ in 5.0 mL of anhydrous dioxane at 25 °C for 17 h under N_2 atmosphere. [b] 10 mol% of $(PhO)_2POOH$ was used. [c] Reaction time is 36 h. [d] The reaction was conducted at 40 °C. [e] Reaction time is 39.5 h.

To demonstrate the practicality of the efficient coupling process, we carried out the reaction for gram scale synthesis of **3ao**: 1.4572 g **3ao** was afforded with 91% yield under standard conditions (eq 1). We studied the reaction of optically active propargylic alcohols[10] with boronic acid **2a** under the standard reaction conditions leading to the formation of (*R*)-**3aa** in 2% ee (eq 2). The Pd(0) would undergo S_N2'-type oxidative addition with propargylic alcohol (*R*)-**1a** in the presence of H^+ in *anti*-stereospecific manner to give intermediate (S_a)-**II**, which would racemize via the η^3-propargylic/allenylpalladium (**II'**) resulting in the formation of **3aa** with low enantioselectivity (Scheme 5b).[11]



A possible catalytic cycle is proposed in Scheme 5a. The reaction of propargylic alcohol with Brønsted acid (HB) would yield the oxonium ion **I**, which would easily undergo oxidative addition reaction with Pd(0) to produce the allenylpalladium intermediate **II**. Here the racemization would occur. Subsequently, intermediate **II** would undergo transmetalation with the boronic acid to form the allenyl aryl palladium intermediate **III**, which would result in the formation of allenes and regeneration of Pd(0) species to complete the catalytic cycle via reductive elimination.

We tried to realize the catalytic asymmetric synthesis of chiral tetrasubstituted allenes. So we screened some chiral ligands at 60 °C (Table 2). 86% of propargylic alcohol **1a** was recovered when we used (*R*)-Ph-Segphos as ligand (Entry 1). Other ligands, such as (*R*)-MeO-BIPHEP, (*R*)-Ph-Garphos and (*R*)-BINAP, also failed to afford the desired allene product **3aa** (Entries 2—4). We are still working on it.

Finally, the potential biological activities of the newly synthesized compounds were evaluated. We first examined the effect of

Scheme 5 The proposed mechanism

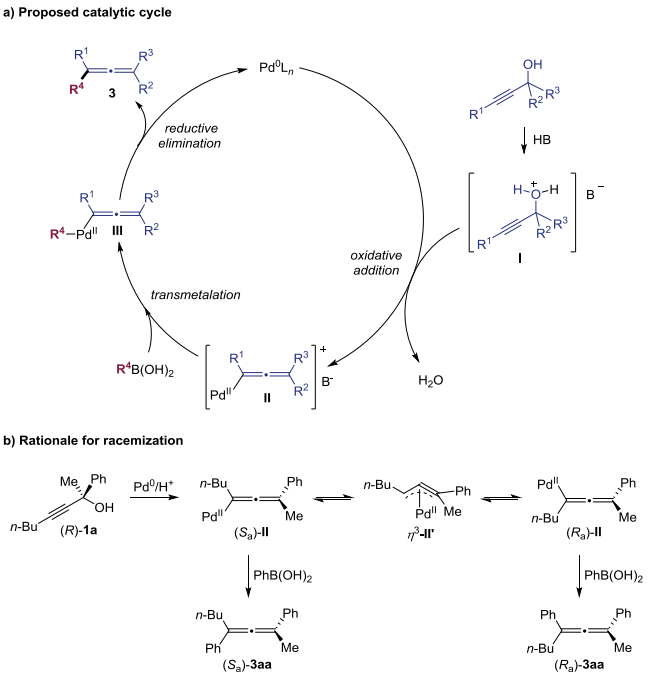
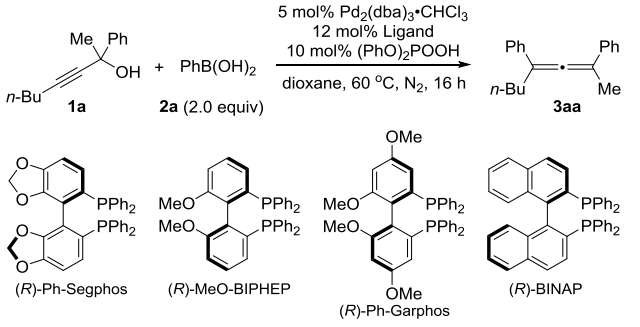


Table 2 Chiral ligand screening[a]



Entry	Ligand	NMR yield of **3aa**/%	ee of **3aa**/%	Recovery of **1a**/%
1	(*R*)-Ph-Segphos	—	—	86
2	(*R*)-MeO-BIPHEP	—	—	83
3	(*R*)-Ph-Garphos	—	—	89
4	(*R*)-BINAP	—	—	61

[a] The reaction was conducted with 0.1 mmol of **1a**, 0.2 mmol of **2a**, 5 mol% of $Pd_2(dba)_3$•$CHCl_3$, 12 mol% of Ligand, and 10 mol% of $(PhO)_2POOH$ in 1.0 mL of anhydrous dioxane at 60 °C for 16 h under N_2 atmosphere. The NMR yield and recovery were determined by ^1H NMR analysis.

the compounds on the secretion of glucagon-like peptide-1 (GLP-1), which is an attractive therapeutic target for type II diabetes. As shown in Table 3, **3aj**, **3da** and **3ha** markedly stimulated GLP-1 release in mouse intestinal cell line STC-1 at 100 μmol·L^{-1}, which were more effective than the positive control phorbol-12-myristate-13-acetate (PMA, 1 μmol·L^{-1}). Then the three compounds were further tested and the half maximal effective concentration (EC_{50}) was measured. Notably, **3da** possessed the most potent stimulatory activity with an EC_{50} of 1.6 μmol·L^{-1}, suggesting that it might be a promising anti-diabetic agent for further study.

The anti-diabetic effect of the compounds was also evaluated by using another cell model. Since diabetic patients often have defect in glucose consumption by the peripheral tissues such as skeletal muscle, we used rat L6 myotube cells to investigate the effect of the compounds on the utilization of glucose. Intriguingly,

 © 2020 SIOC, CAS, Shanghai, & WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim

16147065, 2020, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cjoc.201900442 by Zhejiang University, Wiley Online Library on [01/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

3ab, **3ad**, **3ae**, **3aj**, **3an**, **3ao**, **3da** and **3ea** effectively stimulated glucose consumption at 100 μmol·L^{-1}, leading to more than 1.5-fold improvement over the vehicle control DMSO. While lowering the concentration to 30 μmol·L^{-1}, **3ad**, **3an** and **3ao** still exhibited good activity (> 1.2-fold). Further investigations of their anti-diabetic activities and mechanisms are under way.

Table 3 Stimulation of GLP-1 release in STC-1 cells

Compound	Relative activity[a]/%	EC_{50}[b]/(μmol·L^{-1})
3aa	46.1	
3ab	66.5	
3ad	51.2	
3ae	44.5	
3aj	103.5	24.5±7.4
3al	5.9	
3an	51.5	
3ao	5.0	
3aq	80.1	
3ar	74.9	
3as	60.0	
3ba	37.9	
3ca	1.3	
3da	107.4	1.6±0.0
3ea	76.8	
3ha	127.8	> 50

[a] The concentration of the tested compounds was 100 μmol·L^{-1}. The level of GLP-1 was normalized to that in the presence of the positive control PMA (1 μmol·L^{-1}). [b] EC_{50}s are presented as mean ± SD.

Table 4 Stimulation of glucose consumption in rat L6 myotube cells

Compound	Relative activity[a]/%	
	30 μmol·L^{-1}	100 μmol·L^{-1}
3aa	100.7±2.2	127.2±7.6
3ab	119.8±3.3	161.5±8.0
3ad	123.2±1.3	168.4±6.1
3ae	108.1±2.3	157.9±4.0
3aj	104.3±6.6	157.6±4.4
3al	113.6±3.7	131.5±5.0
3an	120.9±2.5	170.5±3.2
3ao	120.4±1.1	151.7±5.1
3aq	109.8±4.0	139.6±2.0
3ar	101.5±6.2	103.0±1.2
3as	105.6±1.5	147.7±1.5
3ba	105.7±4.7	137.8±6.3
3ca	97.0±4.0	101.7±6.2
3da	102.7±5.9	153.1±1.9
3ea	108.1±5.1	155.0±3.0
3ha	105.3±2.9	96.1±4.3
DMSO[b]	100.0±5.5	
Metformin[c]	148.3±3.2	

[a] Data were normalized to that of the DMSO group and presented as mean ± SD. [b] The concentration of DMSO was 1%. [c] The concentration of metformin was 10 mmol·L^{-1}.

Conclusions

In conclusion, we have developed a general and highly efficient catalytic system for the coupling of propargylic alcohols with boronic acids. Many functional groups are compatible under the very mild reaction conditions to afford trisubstituted and tetrasubstituted allenes facilely. Several newly synthesized compounds in this study displayed potent anti-diabetic activities. Further studies on the details of the reaction mechanism as well as the asymmetric protocol are in progress in our laboratory.

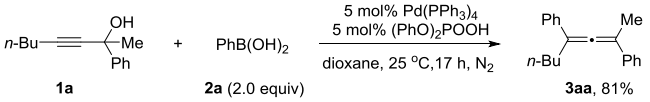
Experimental

General information

NMR spectra were taken with an Agilent-400 spectrometer (400 MHz for ^{1}H NMR, 100 MHz for ^{13}C NMR, and 376 MHz for ^{19}F NMR) in $CDCl_3$. Chemical shifts were recorded in ppm relative to the TMS in $CDCl_3$ and coupling constants were reported in Hz. All reactions were carried out in vial. $Pd(PPh_3)_4$ was purchased from J&K Chemicals and Chembee; $PhB(OH)_2$ was purchased from Bokachem Reagent Co; petroleum ether (b.p. 60—90 °C) was purchased from Shanghai Titan Scientific Co., Ltd. 1,4-Dioxane was dried over sodium wire with benzophenone as the indicator and distilled freshly before use. All the temperatures are referred to the Reaction Block used. NMR yields of allenes were determined by ^{1}H NMR analysis using dibromomethane as the internal standard. The starting propargylic alcohols were synthesized according to the reported procedures.[12]

Synthesis of allenes

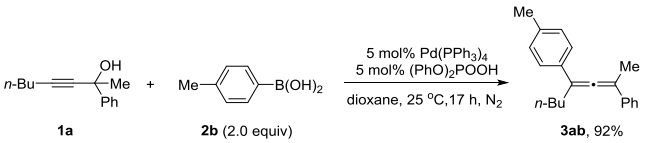
Preparation of 2.4-diphenyl-2,3-octadiene (3aa, QAN-2-149).



1a **2a** (2.0 equiv) **3aa**, 81%

Typical Procedure: To an oven-dried 20 mL vial equipped with a stirring bar were added phenylboronic acid (243.9 mg, 2.0 mmol) and $(PhO)_2POOH$ (12.7 mg, 0.05 mmol). Then the vial was transferred to a glovebox with nitrogen atmosphere. After adding $Pd(PPh_3)_4$ (57.7 mg, 0.05 mmol), **1a** (202.6 mg, 1.0 mmol), and anhydrous dioxane (5 mL), the vial was capped with a Teflon lined lid, sealed with electrical tape, and transferred out of the glovebox. After stirring for 17 h in the reaction block at 25 °C, the reaction mixture was filtered through a short column silica gel (3 cm) eluted with ethyl acetate (20 mL). After removal of the solvent under vacuum, the crude product was analyzed by ^{1}H NMR with CH_2Br_2 (35 μL) as the internal standard: 77% NMR yield of **3aa** was formed based on ^{1}H NMR analysis. The residue was purified by column chromatography on silica gel to afford **3aa** (216.8 mg, 81%, 99% purity) as an oil [eluent: petroleum ether (200 mL)]; ^{1}H NMR (400 MHz, $CDCl_3$) δ: 7.43 (t, J = 7.4 Hz, 4H, Ar-H), 7.34—7.25 (m, 4H, Ar-H), 7.23—7.16 (m, 2H, Ar-H), 2.55 (t, J = 7.6 Hz, 2H, CH_2), 2.20 (s, 3H, CH_3), 1.57 (quintet, J = 7.6 Hz, 2H, CH_2), 1.48—1.34 (m, 2H, CH_2), 0.91 (t, J = 7.2 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) δ: 205.6, 137.3, 137.1, 128.41, 128.38, 126.7, 126.6, 126.1, 125.6, 107.8, 103.6, 30.1, 30.0, 22.6, 16.8, 14.0; MS (70 eV, EI) *m/z* (%): 263 (M^++1, 1.60), 262 (M^+, 7.01), 205 (100); IR (neat) *v*: 3026, 2955, 2925, 2859 1934, 1596, 1490, 1443, 1372, 1065, 1025 cm^{-1}; HRMS calcd for $C_{20}H_{22}$ [M^+]: 262.1722; found 262.1724.

Preparation of 2-phenyl-4-(4-methylphenyl)-2,3-octadiene (3ab, QAN-5-120).



1a **2b** (2.0 equiv) **3ab**, 92%

Following **Typical Procedure**, the reaction of $Pd(PPh_3)_4$ (57.6 mg, 0.05 mmol), *p*-tolylboronic acid (271.6 mg, 2.0 mmol), $(PhO)_2POOH$ (12.5 mg, 0.05 mmol), **1a** (202.3 mg, 1.0 mmol), and anhydrous dioxane (5 mL) at 25 °C afforded **3ab** (255.0 mg, 92%) as an oil [eluent: petroleum ether (200 mL)]; ^{1}H NMR (400 MHz, $CDCl_3$) δ: 7.44 (d, J = 8.0 Hz, 2H, Ar-H), 7.34—7.24 (m, 4H, Ar-H), 7.23—7.16 (m, 1H, Ar-H), 7.11 (d, J = 7.6 Hz, 2H, Ar-H), 2.52 (t, J = 7.4 Hz, 2H, CH_2), 2.32 (s, 3H, CH_3), 2.19 (s, 3H, CH_3), 1.55 (quintet, J = 7.4 Hz, 2H, CH_2), 1.45—1.33 (m, 2H, CH_2), 0.90 (t, J = 7.4 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) δ: 205.3, 137.4, 136.4, 134.0,

 © 2020 SIOC, CAS, Shanghai, & WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim

16147065, 2020, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cjoc.201900442 by Zhejiang University, Wiley Online Library on [01/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

129.1, 128.3, 126.5, 126.0, 125.6, 107.6, 103.4, 30.1, 30.0, 22.6, 21.0, 16.9, 14.0; MS (70 eV, EI) *m/z* (%): 277 (M^++1, 3.31), 276 (M^+, 13.79), 219 (100); **IR** (neat) *v*: 3023, 2924, 2858, 1931, 1597, 1510, 1443, 1377, 1183, 1109, 1065, 1026 cm^{-1}; HRMS calcd for $C_{21}H_{24}$ [M^+]: 276.1878; found 276.1877.

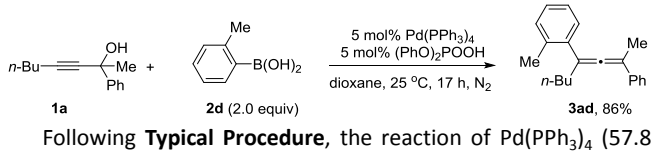
Preparation of 2-phenyl-4-(3-methylphenyl)-2,3-octadiene (3ac, QAN-2-160).



1a **2c** (2.0 equiv) **3ac**, 79%

Following **Typical Procedure**, the reaction of $Pd(PPh_3)_4$ (57.9 mg, 0.05 mmol), *m*-tolylboronic acid (272.3 mg, 2.0 mmol), $(PhO)_2POOH$ (12.6 mg, 0.05 mmol), **1a** (202.3 mg, 1.0 mmol), and anhydrous dioxane (5 mL) at 25 °C afforded **3ac** (232.5 mg, 79%, 94.5% purity) as an oil [eluent: petroleum ether (200 mL)]; ^1H NMR (400 MHz, $CDCl_3$) δ: 7.44 (d, J = 8.0 Hz, 2H, Ar-H), 7.31 (t, J = 7.6 Hz, 2H, Ar-H), 7.26—7.12 (m, 4H, Ar-H), 7.01 (d, J = 7.2 Hz, 1H, Ar-H), 2.62—2.48 (m, 2H, CH_2), 2.32 (s, 3H, CH_3), 2.20 (s, 3H, CH_3), 1.62—1.50 (m, 2H, CH_2), 1.47—1.33 (m, 2H, CH_2), 0.90 (t, J = 7.2 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) δ: 205.5, 137.9, 137.3, 137.0, 128.4, 128.3, 127.5, 126.7, 126.6, 125.6, 123.2, 107.8, 103.4, 30.11, 30.05, 22.6, 21.5, 16.8, 14.0; MS (70 eV, EI) *m/z* (%): 277 (M^++1, 2.50), 276 (M^+, 10.75), 219 (100); IR (neat) *v*: 3024, 2924, 2859, 1933, 1599, 1489, 1444, 1372, 1176, 1094, 1064 cm^{-1}; HRMS calcd for $C_{21}H_{24}$ [M^+]: 276.1878; found 276.1876.

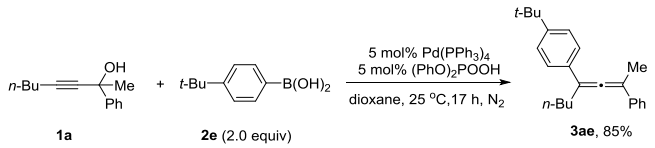
Preparation of 2-phenyl-4-(2-methylphenyl)-2,3-octadiene (3ad, QAN-3-078).



1a **2d** (2.0 equiv) **3ad**, 86%

Following **Typical Procedure**, the reaction of $Pd(PPh_3)_4$ (57.8 mg, 0.05 mmol), *o*-tolylboronic acid (271.9 mg, 2.0 mmol), $(PhO)_2POOH$ (12.6 mg, 0.05 mmol), **1a** (201.9 mg, 1.0 mmol), and anhydrous dioxane (5 mL) at 25 °C afforded **3ad** (236.5 mg, 86%) as an oil [eluent: petroleum ether (200 mL)]; ^1H NMR (400 MHz, $CDCl_3$) δ: 7.45—7.38 (m, 2H, Ar-H), 7.34—7.26 (m, 3H, Ar-H), 7.20—7.09 (m, 4H, Ar-H), 2.54—2.33 (m, 5H, CH_2 and CH_3), 2.15 (s, 3H, CH_3), 1.55—1.44 (m, 2H, CH_2), 1.43—1.33 (m, 2H, CH_2), 0.89 (t, J = 7.2 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) δ: 203.3, 138.1, 137.8, 135.8, 130.4, 128.2, 128.1, 126.7, 126.3, 125.74, 125.71, 106.7, 100.7, 34.0, 30.1, 22.5, 20.7, 17.1, 14.0; MS (70 eV, EI) *m/z* (%): 277 (M^++1, 1.60), 276 (M^+, 6.31), 219 (100); IR (neat) *v*: 3059, 2924, 2860, 1941, 1799, 1596, 1488, 1449, 1373, 1102, 1064, 1027 cm^{-1}; HRMS calcd for $C_{21}H_{24}$ [M^+]: 276.1878; found 276.1879.

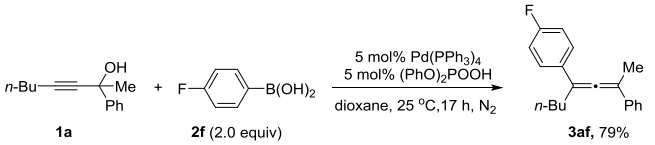
Preparation of 2-phenyl-4-(4-*tert*-butylphenyl)-2,3-octadiene (3ae, QAN-2-164).



1a **2e** (2.0 equiv) **3ae**, 85%

Following **Typical Procedure**, the reaction of $Pd(PPh_3)_4$ (57.8 mg, 0.05 mmol), (4-(*tert*-butyl)phenyl)boronic acid (356.3 mg, 2.0 mmol), $(PhO)_2POOH$ (12.6 mg, 0.05 mmol), **1a** (202.0 mg, 1.0 mmol), and anhydrous dioxane (5 mL) at 25 °C afforded **3ae** (270.6 mg, 85%) as a white solid [eluent: petroleum ether (200 mL)]; m.p. 64.3—68.4 °C (petroleum ether); ^1H NMR (400 MHz, $CDCl_3$) δ: 7.48—7.42 (m, 2H, Ar-H), 7.40—7.27 (m, 6H, Ar-H), 7.24—7.15 (m, 1H, Ar-H), 2.53 (t, J = 7.6 Hz, 2H, CH_2), 2.19 (s, 3H, CH_3), 1.65—1.54 (m, 2H, CH_2), 1.50—1.38 (m, 2H, CH_2), 1.31 (s, 9H, 3 x CH_3), 0.90 (t, J = 7.2 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) δ: 205.4, 149.7, 137.4, 134.0, 128.3, 126.5, 125.7, 125.6, 125.3, 107.5, 103.5, 34.4, 31.3, 30.1, 29.9, 22.7, 16.9, 14.0; MS (70 eV, EI) *m/z* (%): 318 (M^+, 2.39), 261 (100); IR (neat) *v*: 3033, 2954, 2923, 2860, 1928, 1713, 1602, 1494, 1437, 1271, 1180, 1106 cm^{-1}. Anal. calcd for $C_{24}H_{30}$ (%): C 90.51, H 9.49; found C 90.53, H 9.67.

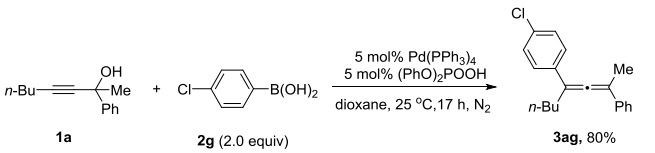
Preparation of 2-phenyl-4-(4-fluorophenyl)-2,3-octadiene (3af, QAN-5-070).



1a **2f** (2.0 equiv) **3af**, 79%

Following **Typical Procedure**, the reaction of $Pd(PPh_3)_4$ (57.7 mg, 0.05 mmol), (4-fluorophenyl)boronic acid (280.2 mg, 2.0 mmol), $(PhO)_2POOH$ (12.5 mg, 0.05 mmol), **1a** (201.7 mg, 1.0 mmol), and anhydrous dioxane (5 mL) at 25 °C afforded **3af** (220.8 mg, 79%) as an oil [eluent: petroleum ether (400 mL)]; ^1H NMR (400 MHz, $CDCl_3$) δ: 7.43 (d, J = 7.6 Hz, 2H, Ar-H), 7.39—7.26 (m, 4H, Ar-H), 7.20 (t, J = 7.2 Hz, 1H, Ar-H), 6.98 (t, J = 8.6 Hz, 2H, Ar-H), 2.58—2.42 (m, 2H, CH_2), 2.20 (s, 3H, CH_3), 1.58—1.50 (m, 2H, CH_2), 1.46—1.35 (m, 2H, CH_2), 0.91 (t, J = 7.2 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) δ: 205.2 (d, J = 1.5 Hz), 161.8 (d, J = 244.9 Hz), 137.1, 133.0 (d, J = 3.2 Hz), 128.4, 127.5 (d, J = 7.9 Hz), 126.7, 125.6, 115.2 (J = 21.4 Hz), 107.0, 103.8, 30.2, 30.0, 22.6, 16.8, 13.9; ^{19}F NMR (376 MHz, $CDCl_3$) δ: –116.7; MS (70 eV, EI) *m/z* (%): 280 (M^+, 4.63), 190 (100); IR (neat) *v*: 3046, 2956, 2928, 2859, 1934, 1600, 1506, 1464, 1227, 1158, 1065 cm^{-1}; HRMS calcd *m/z* for $C_{20}H_{21}F$ [M^+]: 280.1622; found 280.1620.

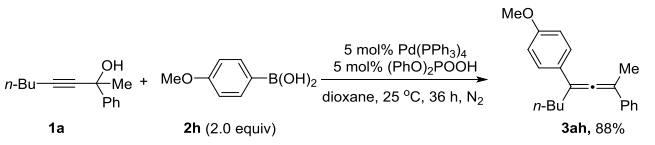
Preparation of 2-phenyl-4-(4-chlorophenyl)-2,3-octadiene (3ag, QAN-2-159, cfsy-3-181).



1a **2g** (2.0 equiv) **3ag**, 80%

Following **Typical Procedure**, the reaction of $Pd(PPh_3)_4$ (57.9 mg, 0.05 mmol), (4-chlorophenyl)boronic acid (312.7 mg, 2.0 mmol), $(PhO)_2POOH$ (12.6 mg, 0.05 mmol), **1a** (201.3 mg, 1.0 mmol), and anhydrous dioxane (5 mL) at 25 °C afforded **3ag** (259.5 mg, 80%, 91% purity) as an oil [eluent: petroleum ether (200 mL)]; ^1H NMR (400 MHz, $CDCl_3$) δ: 7.42 (d, J = 7.2 Hz, 2H, Ar-H), 7.37—7.26 (m, 4H, Ar-H), 7.25—7.13 (m, 3H, Ar-H), 2.59—2.42 (m, 2H, CH_2), 2.20 (s, 3H, CH_3), 1.59—1.48 (m, 2H, CH_2), 1.47—1.33 (m, 2H, CH_2), 0.90 (t, J = 7.4 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) δ: 205.5, 136.9, 135.5, 132.3, 128.5, 128.4, 127.3, 126.8, 125.6, 107.0, 104.0, 30.0, 29.9, 22.6, 16.8, 14.0; MS (70 eV, EI) *m/z* (%): 298 (M^+(^{37}Cl), 3.15), 296 (M^+(^{35}Cl), 9.08), 239 (100); IR (neat) *v*: 3056, 3025, 2959, 2924, 2857, 1929, 1594, 1489, 1462, 1405, 1181, 1090, 1067 cm^{-1}; HRMS calcd for $C_{20}H_{22}{}^{35}Cl$ [M+H^+]: 297.1405; found 297.1403.

Preparation of 2-phenyl-4-(4-methoxylphenyl)-2,3-octadiene (3ah, QAN-2-152).



1a **2h** (2.0 equiv) **3ah**, 88%

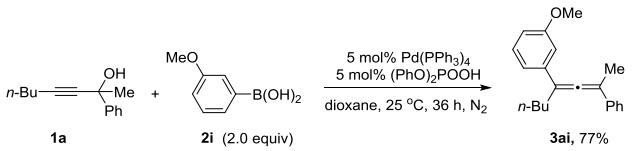
Following **Typical Procedure**, the reaction of $Pd(PPh_3)_4$ (57.9 mg, 0.05 mmol), (4-methoxyphenyl)boronic acid (304.1 mg, 2.0 mmol), $(PhO)_2POOH$ (12.6 mg, 0.05 mmol), **1a** (201.8 mg, 1.0 mmol), and anhydrous dioxane (5 mL) at 25 °C afforded **3ah** (269.4 mg, 88%, 95% purity) as an oil [eluent: petroleum ether (150 mL) to petroleum ether/DCM/Et_2O = 200/1/1 (400 mL)]; ^1H NMR (400 MHz, $CDCl_3$) δ: 7.50—7.41 (m, 2H, Ar-H), 7.38—7.25 (m,

 © 2020 SIOC, CAS, Shanghai, & WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim

16147065, 2020, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cjoc.201900442 by Zhejiang University, Wiley Online Library on [01/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

4H, Ar-H), 7.24—7.16 (m, 1H, Ar-H), 6.85 (d, *J* = 8.4 Hz, 2H, Ar-H), 3.79 (s, 3H, CH_3), 2.65—2.42 (m, 2H, CH_2), 2.19 (s, 3H, CH_3), 1.60—1.50 (m, 2H, CH_2), 1.48—1.33 (m, 2H, CH_2), 0.90 (t, *J* = 7.2 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) *δ*: 205.0, 158.5, 137.5, 129.2, 128.3, 127.1, 126.5, 125.6, 113.9, 107.3, 103.4, 55.3, 30.12, 30.08, 22.6, 16.9, 14.0; MS (70 eV, EI) *m/z* (%): 293 (M^++1, 4.92), 292 (M^+, 21.34), 235 (100); IR (neat) *v*: 3082, 3059, 2929, 2836, 1926, 1604, 1508, 1441, 1374, 1176, 1114, 1064 cm^{-1}; HRMS calcd for $C_{21}H_{25}O$ [M+H^+]: 293.1900; found 293.1897.

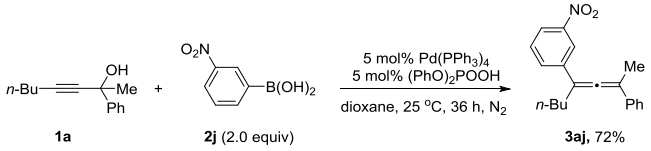
Preparation of 2-phenyl-4-(3-methoxylphenyl)-2,3-octadiene (3ai, QAN-2-161).



1a **2i** (2.0 equiv) **3ai**, 77%

Following **Typical Procedure**, the reaction of $Pd(PPh_3)_4$ (57.7 mg, 0.05 mmol), (3-methoxyphenyl)boronic acid (304.1, 2.0 mmol), $(PhO)_2POOH$ (12.7 mg, 0.05 mmol), **1a** (201.8 mg, 1.0 mmol), and anhydrous dioxane (5 mL) at 25 °C afforded **3ai** (231.8 mg, 77%, 97% purity) as an oil [eluent: petroleum ether (400 mL); ^{1}H NMR (400 MHz, $CDCl_3$) *δ*: 7.44 (d, *J* = 8.0 Hz, 2H, Ar-H), 7.31 (t, *J* = 7.8 Hz, 2H, Ar-H), 7.27—7.12 (m, 2H, Ar-H), 7.03 (d, *J* = 7.6 Hz, 1H, Ar-H), 7.00—6.96 (m, 1H, Ar-H), 6.75 (dd, J_1= 8.2 Hz, J_2= 2.2 Hz, 1H, Ar-H), 3.78 (s, 3H, CH_3), 2.62—2.42 (m, 2H, CH_2), 2.20 (s, 3H, CH_3), 1.64—1.50 (m, 2H, CH_2), 1.46—1.33 (m, 2H, CH_2), 0.90 (t, *J* = 7.4 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) *δ*: 205.6, 159.7, 138.7, 137.2, 129.3, 128.4, 126.6, 125.6, 118.7, 112.1, 111.7, 107.7, 103.6, 55.2, 30.1, 30.0, 22.6, 16.8, 14.0; MS (70 eV, EI) *m/z* (%): 293 (M^++1, 5.98), 292 (M^+, 25.45), 235 (100); IR (neat) *v*: 3059, 2957, 2929, 2836, 1925, 1604, 1580, 1440, 1374, 1240, 1176, 1114, 1040 cm^{-1}; HRMS calcd *m/z* for $C_{21}H_{24}O$ [M^+]: 292.1822; found 292.1827.

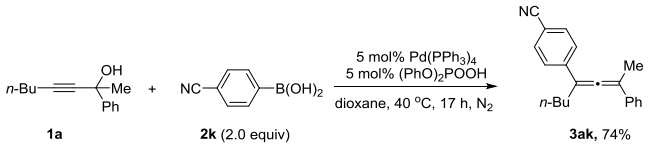
Preparation of 2-phenyl-4-(3-nitrophenyl)-2,3-octadiene (3aj, QAN-2-150).



1a **2j** (2.0 equiv) **3aj**, 72%

Following **Typical Procedure**, the reaction of $Pd(PPh_3)_4$ (57.9 mg, 0.05 mmol), (3-nitrophenyl)boronic acid (334.0 mg, 2.0 mmol), $(PhO)_2POOH$ (12.7 mg, 0.05 mmol), **1a** (202.5 mg, 1.0 mmol), and anhydrous dioxane (5 mL) at 25 °C afforded **3aj** (242.2 mg, 72%, 91% purity) as an oil [eluent: petroleum ether (150 mL) to petroleum ether/DCM/Et_2O = 100/1/1 (~400 mL)]; ^{1}H NMR (400 MHz, $CDCl_3$) *δ*: 8.23 (s, 1H, Ar-H), 8.02 (d, *J* = 8.0 Hz, 1H, Ar-H), 7.73 (d, *J* = 8.0 Hz, 1H, Ar-H), 7.42 (t, *J* = 7.4 Hz, 3H, Ar-H), 7.33 (t, *J* = 7.4 Hz, 2H, Ar-H), 7.23 (t, *J* = 6.4 Hz, 2H, Ar-H), 2.70—2.47 (m, 2H, CH_2), 2.25 (s, 3H, CH_3), 1.58 (quintet, *J* = 7.2 Hz, 2H, CH_2), 1.50—1.33 (m, 2H, CH_2), 0.93 (t, *J* = 7.4 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) *δ*: 205.9, 148.6, 139.3, 136.2, 132.2, 129.1, 128.5, 127.1, 125.7, 121.4, 120.3, 106.5, 105.1, 29.83, 29.80, 22.5, 16.7, 13.9; MS (70 eV, EI) *m/z* (%): 308 (M^++1, 2.72), 307 (M^+, 11.84), 250 (100); IR (neat) *v*: 3084, 2956, 2926, 2861, 1933, 1598, 1493, 1443, 1344, 1067, 1025 cm^{-1}; HRMS calcd for $C_{20}H_{21}NO_2$ [M^+]: 307.1572; found 307.1574.

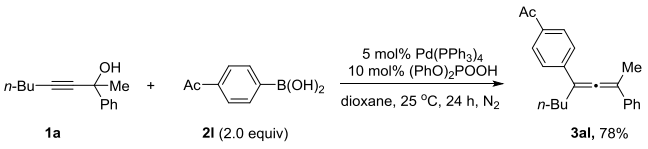
Preparation of 2-phenyl-4-(4-cyanophenyl)-2,3-octadiene (3ak, QAN-3-031).



1a **2k** (2.0 equiv) **3ak**, 74%

Following **Typical Procedure**, the reaction of $Pd(PPh_3)_4$ (57.6 mg, 0.05 mmol), (4-cyanophenyl)boronic acid (294.0, 2.0 mmol), $(PhO)_2POOH$ (12.8 mg, 0.05 mmol), **1a** (202.0 mg, 1.0 mmol), and anhydrous dioxane (5 mL) at 40 °C afforded **3ak** (212.7 mg, 74%) as an oil [eluent: petroleum ether (200 mL) to petroleum ether/DCM/Et_2O = 100/1/1 (400 mL) to petroleum ether/DCM/Et_2O = 50/1/1 (300 mL)]; ^{1}H NMR (400 MHz, $CDCl_3$) *δ*: 7.57 (d, *J* = 8.8 Hz, 2H, Ar-H), 7.49 (d, *J* = 8.8 Hz, 2H, Ar-H), 7.44—7.37 (m, 2H, Ar-H), 7.34 (t, *J* = 7.6 Hz, 2H, Ar-H), 727—7.21 (m, 1H, Ar-H), 2.60—2.45 (m, 2H, CH_2), 2.22 (s, 3H, CH_3), 1.63—1.50 (m, 2H, CH_2), 1.48—1.36 (m, 2H, CH_2), 0.92 (t, *J* = 7.4 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) *δ*: 206.8, 142.2, 136.1, 132.2, 128.5, 127.1, 126.5, 125.6, 119.1, 109.8, 107.1, 104.8, 29.9, 29.6, 22.5, 16.6, 13.9; MS (70 eV, EI) *m/z* (%): 288 (M^++1, 1.12), 287 (M^+, 4.60), 230 (100); IR (neat) *v*: 3058, 2924, 2851, 2226, 1926, 1598, 1496, 1308, 1182, 1064, 1024 cm^{-1}; HRMS calcd *m/z* for $C_{21}H_{21}N$ [M^+]: 287.1669; found 287.1661.

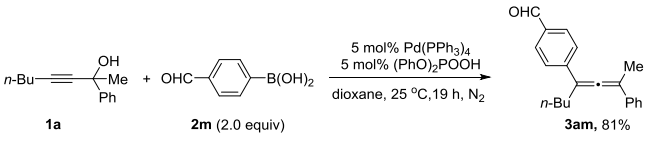
Preparation of 2-phenyl-4-(4-acetylphenyl)-2,3-octadiene (3al, QAN-3-039).



1a **2l** (2.0 equiv) **3al**, 78%

Following **Typical Procedure**, the reaction of $Pd(PPh_3)_4$ (57.8 mg, 0.05 mmol), (4-acetylphenyl)boronic acid (327.8 mg, 2.0 mmol), $(PhO)_2POOH$ (25.2 mg, 0.10 mmol), **1a** (202.1 mg, 1.0 mmol), and anhydrous dioxane (5 mL) at 25 °C afforded **3al** (237.0 mg, 78%) as an oil [eluent: petroleum ether/DCM/Et_2O = 100/1/1 (200 mL) to petroleum ether/DCM/Et_2O = 40/1/1 (~400 mL)]; ^{1}H NMR (400 MHz, $CDCl_3$) *δ*: 7.89 (d, *J* = 8.4 Hz, 2H, Ar-H), 7.50 (d, *J* = 8.0 Hz, 2H, Ar-H), 7.43 (d, *J* = 7.6 Hz, 2H, Ar-H), 7.33 (t, *J* = 7.6 Hz, 2H, Ar-H), 7.27—7.18 (m, 1H, Ar-H), 2.62—2.46 (m, 5H, CH_2 and CH_3), 2.23 (s, 3H, CH_3), 1.57 (quintet, *J* = 7.3 Hz, 2H, CH_2), 1.49—1.35 (m, 2H, CH_2), 0.92 (t, *J* = 7.2 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) *δ*: 206.8, 197.5, 142.2, 136.5, 135.3, 128.55, 128.46, 126.9, 126.0, 125.7, 107.5, 104.3, 30.0, 29.8, 26.5, 22.5, 16.7, 13.9; MS (70 eV, EI) *m/z* (%): 305 (M^++1, 6.24), 304 (M^+, 26.17), 43 (100); IR (neat) *v*: 3346, 3081, 2958, 2924, 2858, 1926, 1680, 1596, 1492, 1410, 1358, 1260, 1183, 1021 cm^{-1}; HRMS calcd for $C_{22}H_{24}O$ [M^+]: 304.1827; found 304.1831.

Preparation of 2-phenyl-4-(4-formylphenyl)-2,3-octadiene (3am, QAN-3-122).



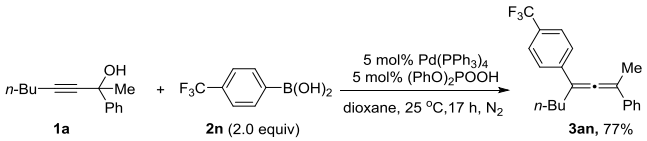
1a **2m** (2.0 equiv) **3am**, 81%

Following **Typical Procedure**, the reaction of $Pd(PPh_3)_4$ (57.7 mg, 0.05 mmol), (4-formylphenyl)boronic acid (300.1 mg, 2.0 mmol), $(PhO)_2POOH$ (12.7 mg, 0.05 mmol), **1a** (202.3 mg, 1.0 mmol), and anhydrous dioxane (5 mL) at 25 °C afforded **3am** (236.4 mg, 81%) as a white solid [eluent: petroleum ether (200 mL) to petroleum ether/DCM/Et_2O = 50/1/1 (~410 mL)]; m.p. 56.3—59.4 °C; ^{1}H NMR (400 MHz, $CDCl_3$) *δ*: 9.96 (s, 1H, CHO), 7.81 (d, *J* = 8.0 Hz, 2H, Ar-H), 7.57 (d, *J* = 8.4 Hz, 2H, Ar-H), 7.45—7.39 (m, 2H, Ar-H), 7.34 (t, *J* = 7.6 Hz, 2H, Ar-H), 7.26—7.20 (m, 1H, Ar-H), 2.60—2.54 (m, 2H, CH_2), 2.24 (s, 3H, CH_3), 1.63—1.50 (m, 2H, CH_2), 1.49—1.39 (m, 2H, CH_2), 0.92 (t, *J* = 7.4 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) *δ*: 207.2, 191.7, 143.8, 136.4, 134.7, 129.9, 128.5, 127.0, 126.4, 125.7, 107.6, 104.5, 30.0, 29.8, 22.5, 16.7, 13.9; MS (70 eV, EI) *m/z* (%): 291 (M^++1, 3.75), 290 (M^+, 15.17), 205 (100); IR (neat) *v*: 3356, 3083, 2925, 2854, 1924, 1694, 1596, 1491, 1459, 1304, 1208, 1110, 1064 cm^{-1}. Anal. calcd for $C_{21}H_{22}O$ (%): C 86.85, H 7.64; found C 86.66, H 7.69.

 © 2020 SIOC, CAS, Shanghai, & WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim

16147065, 2020, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cjoc.201900442 by Zhejiang University, Wiley Online Library on [01/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

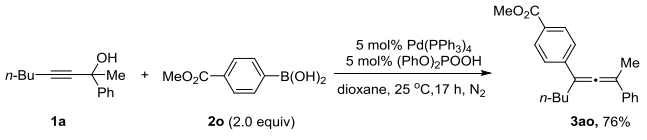
Preparation of 2-phenyl-4-(4-trifluoromethylphenyl)-2,3-octadiene (3an, QAN-2-157).



1a **2n** (2.0 equiv) **3an**, 77%

Following **Typical Procedure**, the reaction of $Pd(PPh_3)_4$ (57.8 mg, 0.05 mmol), (4-(trifluoromethyl)phenyl)boronic acid (378.0 mg, 2.0 mmol), $(PhO)_2POOH$ (12.6 mg, 0.05 mmol), **1a** (202.5 mg, 1.0 mmol), and anhydrous dioxane (5 mL) at 25 °C afforded **3an** (282.0 mg, 77%, 91% purity) as an oil [eluent: petroleum ether (200 mL)]; ^1H NMR (400 MHz, $CDCl_3$) δ: 7.59—7.45 (m, 4H, Ar-H), 7.43 (d, J = 7.6 Hz, 2H, Ar-H), 7.33 (t, J = 7.6 Hz, 2H, Ar-H), 7.23 (t, J = 7.2 Hz, 1H, Ar-H), 2.63—2.44 (m, 2H, CH_2), 2.22 (s, 3H, CH_3), 1.63—1.51 (m, 2H, CH_2), 1.49—1.33 (m, 2H, CH_2), 0.92 (t, J = 7.2 Hz, 3H, CH_3); ^{19}F NMR (376 MHz, $CDCl_3$) δ: –62.9; ^{13}C NMR (100 MHz, $CDCl_3$) δ: 206.3, 140.9, 136.6, 128.54 (q, J = 32.1 Hz), 128.51, 127.0, 126.2, 125.7, 125.3 (q, J = 3.8 Hz), 124.3 (q, J = 270.2 Hz), 107.1, 104.4, 30.0, 29.8, 22.6, 16.7, 13.9; MS (70 eV, EI) *m/z* (%): 331 (M^++1, 2.52), 330 (M^+, 10.68), 273 (100); IR (neat) *v*: 3082, 2957, 2933, 2866, 1927, 1613, 1463, 1375, 1323, 1156, 1113, 1066 cm^{-1}; HRMS calcd for $C_{21}H_{21}F_3$ [M^+]: 330.1595; found 330.1597.

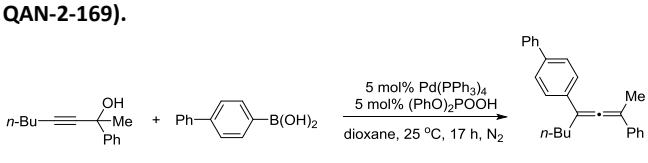
Preparation of 2-phenyl-4-(4-methoxylcarbonylphenyl)-2,3-octadiene (3ao, QAN-2-145).



1a **2o** (2.0 equiv) **3ao**, 76%

Following **Typical Procedure**, the reaction of $Pd(PPh_3)_4$ (57.8 mg, 0.05 mmol), (4-(methoxycarbonyl)phenyl)boronic acid (360.1 mg, 2.0 mmol), $(PhO)_2POOH$ (12.6 mg, 0.05 mmol), **1a** (202.1 mg, 1.0 mmol), and anhydrous dioxane (5 mL) at 25 °C afforded **3ao** (246.1 mg, 76%, 99% purity) as a white solid [eluent: petroleum ether (200 mL) to petroleum ether/DCM/Et_2O = 100/1/1 (200 mL)]; m.p. 65.7—70.8 °C (petroleum ether/dichloromethane); ^1H NMR (400 MHz, $CDCl_3$) δ: 7.96 (d, J = 8.4 Hz, 2H, Ar-H), 7.52—7.38 (m, 4H, Ar-H), 7.33 (t, J = 7.6 Hz, 2H, Ar-H), 7.22—7.16 (m, 1H, Ar-H), 3.90 (s, 3H, CH_3), 2.63—2.50 (m, 2H, CH_2), 2.22 (s, 3H, CH_3), 1.62—1.43 (m, 2H, CH_2), 1.50—1.32 (m, 2H, CH_2), 0.91 (t, J = 7.4 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) δ: 206.6, 166.9, 142.0, 136.6, 129.7, 128.4, 128.1, 126.9, 125.8, 125.6, 107.5, 104.2, 52.0, 30.0, 29.8, 22.5, 16.7, 13.9; MS (70 eV, EI) *m/z* (%): 321 (M^++1, 3.04), 320 (M^+, 12.24), 263 (100); IR (neat) *v*: 3033, 2954, 2923, 2860, 1928, 1712, 1602, 1494, 1372, 1271, 1180, 1106, 1019 cm^{-1}. Anal. calcd for $C_{22}H_{24}O_2$ (%): C 82.46, H 7.55; found C 82.37, H 7.62.

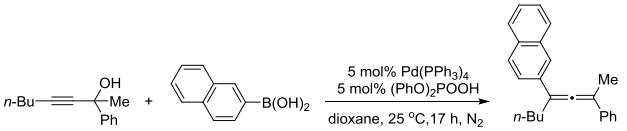
Preparation of 2-phenyl-4-biphenyl-2,3-octadiene (3ap, QAN-2-169).



1a **2p** (2.0 equiv) **3ap**, 82%

Following **Typical Procedure**, the reaction of $Pd(PPh_3)_4$ (57.7 mg, 0.05 mmol), [1,1'-biphenyl]-4-ylboronic acid (396.3, 2.0 mmol), $(PhO)_2POOH$ (12.6 mg, 0.05 mmol), **1a** (202.3 mg, 1.0 mmol), and anhydrous dioxane (5 mL) at 25 °C afforded **3ap** (302.6 mg, 82%, purity: 92%) as an oil [eluent: petroleum ether (200 mL)]; ^1H NMR (400 MHz, $CDCl_3$) δ: 7.63—7.38 (m, 10H, Ar-H), 7.36—7.27 (m, 3H, Ar-H), 7.25—7.16 (m, 1H, Ar-H), 2.58 (t, J = 7.4 Hz, 2H, CH_2), 2.23 (s, 3H, CH_3), 1.59 (quintet, J = 7.5 Hz, 2H, CH_2), 1.50—1.35 (m, 2H, CH_2), 0.92 (t, J = 7.4 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) δ: 205.8, 140.9, 139.5, 137.2, 136.0, 128.7, 128.4, 127.1, 126.9, 126.7, 126.5, 125.7, 107.5, 103.8, 30.1, 30.0, 22.6, 16.9, 14.0; MS (70 eV, EI) *m/z* (%): 339 (M^++1, 6.58), 338 (M^+, 22.32), 281 (100); IR (neat) *v*: 3055, 3027, 2923, 2857, 1931, 1596, 1486, 1370, 1230, 1111, 1067 cm^{-1}; HRMS calcd *m/z* for $C_{26}H_{26}$ [M^+]: 338.2029; found 338.2027.

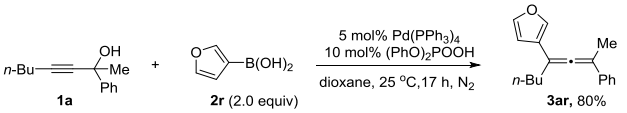
Preparation of 2-phenyl-4-(2-napthyl)-2,3-octadiene (3aq, QAN-2-174).



1a **2q** (2.0 equiv) **3aq**, 88%

Following **Typical Procedure**, the reaction of $Pd(PPh_3)_4$ (57.8 mg, 0.05 mmol), naphthalen-2-ylboronic acid (344.2 mg, 2.0 mmol), $(PhO)_2POOH$ (12.6 mg, 0.05 mmol), **1a** (202.6 mg, 1.0 mmol), and anhydrous dioxane (5 mL) at 25 °C afforded **3aq** (276.0 mg, 88%) as a white solid [eluent: petroleum ether (200 mL)]; m.p. 72.7—79.8 °C; ^1H NMR (400 MHz, $CDCl_3$) δ: 7.87—7.43 (m, 3H, Ar-H), 7.71 (d, J = 8.8 Hz, 1H, Ar-H), 7.63—7.52(m, 1H, Ar-H), 7.50—7.36 (m, 4H, Ar-H), 7.33 (t, J = 7.6 Hz, 2H, Ar-H), 7.28—7.15 (m, 1H, Ar-H), 2.68 (t, J = 7.6 Hz, 2H, CH_2), 2.25 (s, 3H, CH_3), 1.63 (quintet, J = 7.5 Hz, 2H, CH_2), 1.53—1.37 (m, 2H, CH_2), 0.93 (t, J = 7.4 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) δ: 206.3, 137.2, 134.5, 133.7, 132.5, 128.4, 128.0, 127.8, 127.5, 126.7, 126.0, 125.7, 125.6, 125.5, 123.6, 108.1, 103.9, 30.1, 30.0, 22.7, 16.9, 14.0; MS (70 eV, EI) *m/z* (%): 313 (M^++1, 5.92), 312 (M^+, 22.68), 255 (100); IR (neat) *v*: 3055, 2955, 2923, 2866, 1922, 1628, 1595, 1491, 1463, 1261, 1187, 1127, 1064 cm^{-1}; HRMS calcd *m/z* for $C_{24}H_{24}$ [M^+]: 312.1873; found 312.1870.

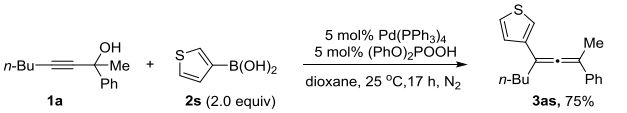
Preparation of 2-phenyl-4-(3-furyl)-2,3-octadiene (3ar, QAN-3-003).



1a **2r** (2.0 equiv) **3ar**, 80%

Following **Typical Procedure**, the reaction of $Pd(PPh_3)_4$ (57.8 mg, 0.05 mmol), furan-3-ylboronic acid (223.9 mg, 2.0 mmol), $(PhO)_2POOH$ (25.2 mg, 0.10 mmol), **1a** (202.4 mg, 1.0 mmol), and anhydrous dioxane (5 mL) at 25 °C afforded **3ar** (201.8 mg, 80%) as an oil [eluent: petroleum ether (300 mL)]; ^1H NMR (400 MHz, $CDCl_3$) δ: 7.48—7.37 (m, 3H, Ar-H), 7.36—7.26 (m, 3H, Ar-H), 7.20 (t, J = 7.4 Hz, 1H, Ar-H), 6.35 (d, J = 0.8 Hz, 1H, Ar-H), 2.48—2.25 (m, 2H, CH_2), 2.17 (s, 3H, CH_3), 1.62—1.50 (m, 2H, CH_2), 1.48—1.31 (m, 2H, CH_2), 0.90 (t, J = 7.2 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) δ: 204.1, 143.1, 138.2, 137.4, 128.3, 126.6, 125.8, 123.5, 109.4, 103.2, 100.6, 30.7, 29.9, 22.6, 17.0, 14.0; MS (70 eV, EI) *m/z* (%): 253 (M^++1, 4.92), 252 (M^+, 15.16), 105 (100); IR (neat) *v*: 3059, 2956, 2927, 2863, 1935, 1762, 1598, 1446, 1265, 1157, 1066 cm^{-1}; HRMS calcd for $C_{18}H_{20}O$ [M^+]: 252.1514; found 252.1510.

Preparation of 2-phenyl-4-(3-thienyl)-2,3-octadiene (3as, QAN-2-196).



1a **2s** (2.0 equiv) **3as**, 75%

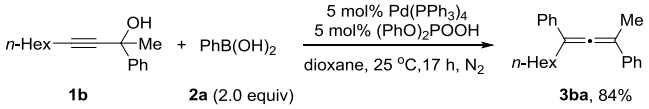
Following **Typical Procedure**, the reaction of $Pd(PPh_3)_4$ (57.8 mg, 0.05 mmol), thiophen-3-ylboronic acid (256.1mg, 2.0 mmol), $(PhO)_2POOH$ (12.6 mg, 0.05 mmol), **1a** (201.9 mg, 1.0 mmol), and anhydrous dioxane (5 mL) at 25 °C afforded **3as** (217.1 mg, 75%, 93% purity) as an oil [eluent: petroleum ether (300 mL)]; ^1H NMR (400 MHz, $CDCl_3$) δ: 7.48—7.38 (m, 2H, Ar-H), 7.34—7.25 (m, 2H, Ar-H), 7.23—7.15 (m, 2H, Ar-H), 7.14—7.03 (m, 2H, Ar-H), 2.32 (dt, J_1 = 7.2 Hz, J_2 = 2.3 Hz, 2H, CH_2), 2.18 (s, 3H, CH_3), 1.57 (quintet, J =

 © 2020 SIOC, CAS, Shanghai, & WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim

16147065, 2020, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cjoc.201900442 by Zhejiang University, Wiley Online Library on [01/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

7.4 Hz, 2H, CH_2), 1.48—1.33 (m, 2H, CH_2), 0.90 (t, *J* = 7.2 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) *δ*: 205.3, 139.0, 137.3, 128.3, 127.0, 126.6, 125.7, 125.2, 118.8, 104.1, 103.2, 30.7, 30.0, 22.6, 17.0, 14.0; MS (70 eV, EI) *m/z* (%): 269 (M^++1, 5.23), 268 (M^+, 24.95), 211 (100); IR (neat) *v*: 3026, 2954, 2924, 2858, 1934, 1597, 1491, 1373, 1233, 1178, 1065 cm^{-1}; HRMS calcd for $C_{18}H_{20}S$ [M^+]: 268.1286; found 268.1289.

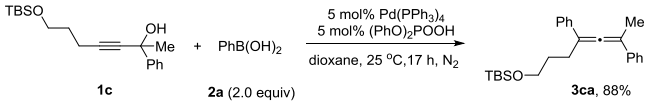
Preparation of 2,4-diphenyl-2,3-decadiene (3ba, QAN-3-090).



1b **2a** (2.0 equiv) **3ba**, 84%

Following **Typical Procedure**, the reaction of $Pd(PPh_3)_4$ (57.8 mg, 0.05 mmol), phenylboronic acid (243.9, 2.0 mmol), $(PhO)_2POOH$ (12.6 mg, 0.05 mmol), **1b** (229.8 mg, 1.0 mmol), and anhydrous dioxane (5 mL) at 25 °C afforded **3ba** (251.1 mg, 84%, 96% purity) as an oil [eluent: petroleum ether (200 mL)]; ^{1}H NMR (400 MHz, $CDCl_3$) *δ*: 7.43 (t, *J* = 7.4 Hz, 4H, Ar-H), 7.34—7.26 (m, 4H, Ar-H), 7.22—7.15 (m, 2H, Ar-H), 2.60—2.45 (m, 2H, CH_2), 2.20 (s, 3H, CH_3), 1.58 (quintet, *J* = 7.4 Hz, 2H, CH_2), 1.45—1.32 (m, 2H, CH_2), 1.29—1.20 (m, 4H, 2 x CH_2), 0.85 (t, *J* = 6.6 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, CDCl3) *δ*: 205.6, 137.3, 137.1 128.40, 128.37, 126.7, 126.6, 126.1, 125.6, 107.8, 103.6, 31.7, 30.2, 29.3, 27.9, 22.7, 16.8, 14.0; MS (70 eV, EI) *m/z* (%): 291 (M^++1, 1.68), 290 (M^+, 6.86), 205 (100); IR (neat) *v*: 3027, 2923, 2855, 1935, 1595, 1490, 1447, 1372, 1181, 1067, 1024 cm^{-1}; HRMS calcd for $C_{22}H_{26}$ [M^+]: 290.2035; found 290.2037.

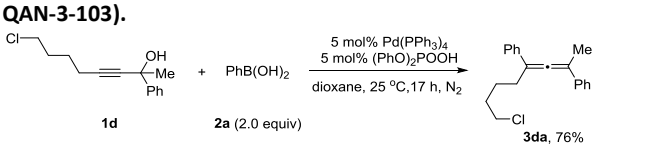
Preparation of 2,4-diphenyl-2,3-heptadiene-7-(*tert*-butyldimethylsilyloxy) ether (3ca, QAN-3-102).



1c **2a** (2.0 equiv) **3ca**, 88%

Following **Typical Procedure**, the reaction of $Pd(PPh_3)_4$ (57.8 mg, 0.05 mmol), phenylboronic acid (244.2 mg, 2.0 mmol), $(PhO)_2POOH$ (12.6 mg, 0.05 mmol), **1c** (318.3 mg, 1.0 mmol), and anhydrous dioxane (5 mL) at 25 °C afforded **3ca** (332.9 mg, 88%) as an oil [eluent: petroleum ether (200 mL) to petroleum ether/DCM/Et_2O = 100/1/1 (400 mL)]; ^{1}H NMR (400 MHz, $CDCl_3$) *δ*: 7.49—7.43 (m, 4H, Ar-H), 7.35—7.28 (m, 4H, Ar-H), 7.24—7.17 (m, 2H, Ar-H), 3.72 (t, *J* = 6.2 Hz, 2H, CH_2), 2.65 (t, *J* = 7.4 Hz, 2H, CH_2), 2.22 (s, 3H, CH_3), 1.90—1.79 (m, 2H, CH_2), 0.91 (s, 9H, 3 x CH_3), 0.05 (s, 6H, 2 x CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) *δ*: 205.4, 137.1, 136.9, 128.42, 128.39, 126.73, 126.69, 126.0, 125.6, 107.5, 104.0, 62.8, 31.2, 26.5, 26.0, 18.3, 16.9, −5.3; MS (ESI) *m/z* (%): 247 (M-OTBS)$^+$, 379 (M+H)$^+$; IR (neat) *v*: 3056, 3028, 2930, 2856, 1933, 1935, 1596, 1491, 1460, 1384, 1251, 1184, 1098 cm^{-1}; HRMS calcd for $C_{25}H_{35}OSi$ [M+H]$^+$: 379.2452; found 379.2448.

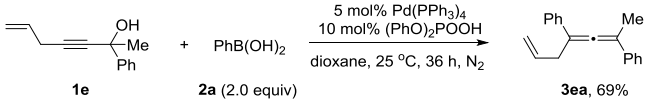
Preparation of 2,4-diphenyl-8-chloro-2,3-octadien (3da, QAN-3-103).



1d **2a** (2.0 equiv) **3da**, 76%

Following **Typical Procedure**, the reaction of $Pd(PPh_3)_4$ (57.8 mg, 0.05 mmol), phenylboronic acid (244.2 mg, 2.0 mmol), $(PhO)_2POOH$ (12.6 mg, 0.05 mmol), **1d** (236.5 mg, 1.0 mmol), and anhydrous dioxane (5 mL) at 25 °C afforded **3da** (234.8 mg, 76%, 96% purity) as an oil [eluent: petroleum ether (500 mL)]; ^{1}H NMR (400 MHz, $CDCl_3$) *δ*: 7.49—7.38 (m, 4H, Ar-H), 7.36—7.21 (m, 4H, Ar-H), 7.25—7.15 (m, 2H, Ar-H), 3.50 (t, *J* = 6.8 Hz, 2H, CH_2), 2.58 (t, *J* = 7.6 Hz, 2H, CH_2), 2.21 (s, 3H, CH_3), 1.92—1.81 (m, 2H, CH_2), 1.78—1.62 (m, 2H, CH_2); ^{13}C NMR (100 MHz, $CDCl_3$) *δ*: 205.4, 136.9, 136.6, 128.5, 128.4, 126.81, 126.77, 126.0, 125.6, 107.1, 104.1, 44.7, 32.3, 29.4, 25.1, 16.9; MS (70 eV, EI) *m/z* (%): 298 (M^+(^{37}Cl), 3.17), 296 (M^+(^{35}Cl), 8.77), 205 (100); IR (neat) *v*: 3078, 2988, 2941, 2853, 1927, 1591, 1444, 1331, 1283, 1209, 1069 cm^{-1}; HRMS calcd for $C_{20}H_{22}{}^{35}Cl$ [M+H$^+$]: 297.1405; found 297.1403.

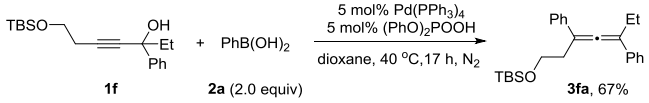
Preparation of 2,4-diphenyl-2,3,6-heptatriene (3ea, QAN-3-107).



1e **2a** (2.0 equiv) **3ea**, 69%

Following **Typical Procedure**, the reaction of $Pd(PPh_3)_4$ (57.9 mg, 0.05 mmol), phenylboronic acid (244.0 mg, 2.0 mmol), $(PhO)_2POOH$ (25.0 mg, 0.10 mmol), **1e** (185.9 mg, 1.0 mmol), and anhydrous dioxane (5 mL) at 25 °C afforded **3ae** (170.0 mg, 69%) as an oil [eluent: petroleum ether (400 mL)]; ^{1}H NMR (400 MHz, $CDCl_3$) *δ*: 7.46—7.40 (m, 4H, Ar-H), 7.34—7.28 (m, 4H, Ar-H), 7.22—7.17 (m, 2H, Ar-H), 6.05—5.90 (m, 1H, =CH), 5.22—5.14 (m, 1H, =CH), 5.08—5.01 (m, 1H, =CH), 3.28—3.34 (m, 2H, CH_2), 2.20 (s, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) *δ*: 206.0, 137.0, 136.4, 135.8, 128.42, 128.39, 126.8, 126.1, 125.7, 116.2, 106.1, 104.0, 34.9, 16.8; MS (70 eV, EI) *m/z* (%): 247 (M^++1, 12.63), 246 (M^+, 59.29), 154 (100); IR (neat) *v*: 3063, 2913, 1941, 1590, 1472, 1438, 1379, 1174, 1079 cm^{-1}; HRMS calcd for $C_{19}H_{18}$ [M^+]: 246.1409; found 246.1406.

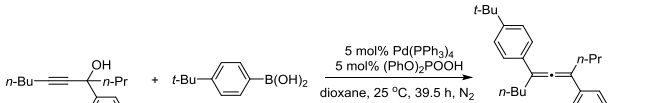
Preparation of 3,5-diphenyl-3,4-heptadiene-7-(*tert*-butyldimethylsilyloxy) ether (3fa, QAN-3-058).



1f **2a** (2.0 equiv) **3fa**, 67%

Following **Typical Procedure**, the reaction of $Pd(PPh_3)_4$ (57.8 mg, 0.05 mmol), phenylboronic acid (243.7 mg, 2.0 mmol), $(PhO)_2POOH$ (12.7 mg, 0.05 mmol), **1f** (318.2 mg, 1.0 mmol), and anhydrous dioxane (5 mL) at 40 °C afforded **3fa** (257.4 mg, 67%, 98% purity) as an oil [eluent: petroleum ether (200 mL) to petroleum ether/DCM/Et_2O = 100/1/1 (200 mL)]; ^{1}H NMR (400 MHz, $CDCl_3$) *δ*: 7.64—7.51 (m, 4H, Ar-H), 7.44 (t, *J* = 7.6 Hz, 4H, Ar-H), 7.37—7.27 (m, 2H, Ar-H), 4.10—3.90 (m, 2H, CH_2), 2.96 (t, *J* = 7.4 Hz, 2H, CH_2), 2.78—2.58 (m, 2H, CH_2), 1.31 (t, *J* = 7.2 Hz, 3H, CH_3), 1.03—0.90 (m, 9H, 3 x CH_3), 0.16 (d, *J* = 3.6 Hz, 6H, 2 x CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) *δ*: 205.0, 136.7, 128.44, 128.42, 128.2, 126.8, 125.9, 125.8, 110.8, 106.5, 62.3, 33.8, 25.94, 25.88, 23.3, 18.3, 12.6, −5.3; MS (70 eV, EI) *m/z* (%): 378 (M^+, 1.25), 349 (100); IR (neat) *v*: 3029, 2955, 2930, 2857, 1934, 1595, 1491, 1460, 1387, 1252, 1096 cm^{-1}; HRMS calcd for $C_{25}H_{34}OSi$ [M^+]: 378.2379; found 378.2385.

Preparation of 4-(3,4-methylenedioxyphenyl)-6-(4-*t*-butylphenyl)-4,5-decadiene (3ge, QAN-3-046).



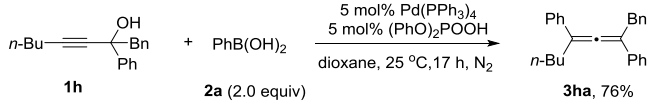
1g **2e** (2.0 equiv) **3ge**, 66%

Following **Typical Procedure,** the reaction of $Pd(PPh_3)_4$ (57.8 mg, 0.05 mmol), (4-(*tert*-butyl)phenyl)boronic acid (356.3 mg, 2.0 mmol), $(PhO)_2POOH$ (12.8 mg, 0.05 mmol), **1g** (274.0 mg, 1.0 mmol), and anhydrous dioxane (5 mL) at 25 °C afforded **3ge** (257.4 mg, 66%) as a white solid [eluent: petroleum ether (200 mL) to petroleum ether/DCM/Et_2O = 100/1/1 (400 mL)]; m.p. 67.2—71.2 °C (petroleum ether/ethyl acetate); ^{1}H NMR (400 MHz, $CDCl_3$) *δ*: 7.45—7.28 (m, 4H, Ar-H), 7.01—6.83 (m, 2H, Ar-H), 6.75 (d, *J* = 8.0 Hz, 1H, Ar-H), 5.98—5.82 (m, 2H, CH_2), 2.59—2.37 (m, 4H, 2 x CH_2), 1.62—1.52 (m, 4H, 2 x CH_2), 1.46—1.35 (m, 2H, CH_2), 1.31 (s, 9H, 3 x CH_3), 0.99 (t, *J* = 7.2 Hz, 3H, CH_3), 0.91 (t, *J* = 7.4 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) *δ*: 204.7, 149.6, 147.8, 146.4, 134.1,

16147065, 2020, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cjoc.201900442 by Zhejiang University, Wiley Online Library on [01/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

131.4, 125.5, 125.3, 118.9, 108.8, 108.7, 108.1, 106.7, 100.9, 34.4, 32.9, 31.3, 30.3, 30.1, 22.8, 21.4, 14.3, 14.0; MS (ESI) *m/z*: 391 ([M+H$^+$]); IR (neat) *v*: 3080, 2956, 2927, 2871, 1926, 1602, 1503, 1464, 1362, 1111, 1086, 1042 cm^{-1}. Anal. calcd for $C_{27}H_{34}O_2$ (%): C 83.03, H 8.77; found C 82.60, H 8.78.

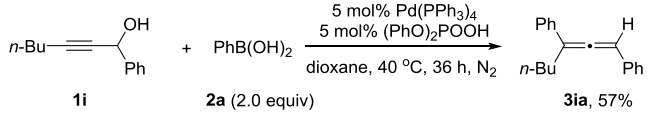
Preparation of 1,3-diphenyl-1-benzyl-2,3-heptadiene (3ha, QAN-3-106).



1h **2a** (2.0 equiv) **3ha**, 76%

Following **Typical Procedure**, the reaction of $Pd(PPh_3)_4$ (57.8 mg, 0.05 mmol), phenylboronic acid (244.1 mg, 2.0 mmol), $(PhO)_2POOH$ (12.7 mg, 0.05 mmol), **1h** (278.9 mg, 1.0 mmol), and anhydrous dioxane (5 mL) at 25 °C afforded **3ha** (257.1 mg, 76%) as an oil [eluent: petroleum ether (600 mL)]; ^{1}H NMR (400 MHz, $CDCl_3$) δ: 7.47—7.41 (m, 2H, Ar-H), 7.35—7.24 (m, 2H, Ar-H), 7.30—7.23 (m, 6H, Ar-H), 7.22—7.10 (m, 5H, Ar-H), 3.90 (s, 2H, CH_2), 2.58—2.38 (m, 2H, CH_2), 1.51—1.41 (m, 2H, CH_3), 1.40—1.30 (m, 2H, CH_2), 0.87 (t, J = 7.2 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) δ: 206.8, 139.4, 136.7, 136.5, 128.9, 128.4, 128.3, 128.2, 126.73, 126.69, 126.11, 126.06, 126.04, 109.4, 107.9, 37.2, 30.21, 30.18, 22.7, 14.0; MS (70 eV, EI) *m/z* (%): 339 (M$^+$+1, 4.70), 338 (M$^+$, 17.22), 91 (100); IR (neat) *v*: 3039, 2935, 2859, 1941, 1591, 1468, 1444, 1375, 1173, 1018 cm^{-1}; HRMS calcd for $C_{26}H_{26}$ [M$^+$]: 338.2035; found 338.2038.

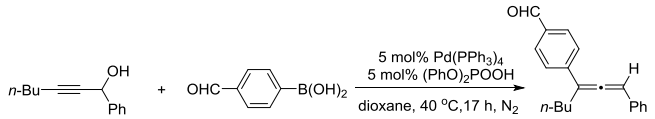
Preparation of 1,3-diphenyl-1,2-heptadiene (3ia, QAN-3-151).



1i **2a** (2.0 equiv) **3ia**, 57%

Following **Typical Procedure**, the reaction of $Pd(PPh_3)_4$ (57.9 mg, 0.05 mmol), phenylboronic acid (243.8 mg, 2.0 mmol), $(PhO)_2POOH$ (12.5 mg, 0.05 mmol), **1i** (188.1 mg, 1.0 mmol), and anhydrous dioxane (5 mL) at 40 °C afforded **3ia**[6h] (142.1 mg, 57%) as an oil [eluent: petroleum ether (200 mL)]; ^{1}H NMR (400 MHz, $CDCl_3$) δ: 7.45 (d, J = 7.6 Hz, 2H, Ar-H), 7.36—7.27 (m, 6H, Ar-H), 7.23—7.16 (m, 2H, Ar-H), 6.52 (t, J = 2.8 Hz, 1H, =CH), 2.63—2.47 (m, 2H, CH_2), 1.66—1.54 (m, 2H, CH_2), 1.48—1.38 (m, 2H, CH_2), 0.92 (t, J = 7.4 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) δ: 206.5, 136.3, 134.7, 128.7, 128.5, 127.0, 126.9, 126.7, 126.1, 110.0, 97.8, 30.1, 29.9, 22.6, 13.9`; MS (70 eV, EI) *m/z* (%): 248 (M$^+$, 4.04), 105 (100); IR (neat) *v*: 3028, 2928, 2870, 1932, 1724, 1597, 1493, 1447, 1378, 1264, 1204, 1071 cm^{-1}.

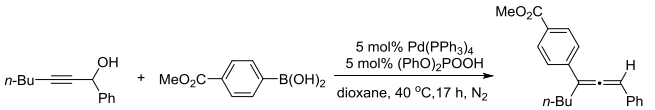
Preparation of 1-phenyl-3-(4-formylphenyl)-1,2-heptadiene (3im, QAN-3-155).



1i **2m** (2.0 equiv) **3im**, 63%

Following **Typical Procedure**, the reaction of $Pd(PPh_3)_4$ (57.7 mg, 0.05 mmol), (4-formylphenyl)boronic acid (299.7 mg, 2.0 mmol), $(PhO)_2POOH$ (12.6 mg, 0.05 mmol), **1i** (188.0 mg, 1.0 mmol), and anhydrous dioxane (5 mL) of 40 °C afforded **3im** (173.6 mg, 63%) as an oil [eluent: petroleum ether/DCM/Et_2O = 100/1/1 (200 mL) to petroleum ether/DCM/Et_2O = 50/1/1 (400 mL)]; ^{1}H NMR (400 MHz, $CDCl_3$) δ: 9.98 (s, 1H, CHO), 7.82 (d, J = 8.4 Hz, 2H, Ar-H), 7.60 (d, J = 8.4 Hz, 2H, Ar-H), 7.36—7.28 (m, 4H, Ar-H), 7.23—7.16 (m, 1H, Ar-H), 6.60 (t, J = 3.0 Hz, 1H, =CH), 2.62—2.52 (m, 2H, CH_2), 1.68—1.56 (m, 2H, CH_2), 1.48—1.40 (m, 2H, CH_2), 0.93 (t, J = 7.4 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) δ: 208.0, 191.7, 142.9, 135.0, 133.8, 129.9, 128.8, 127.4, 126.9, 126.6, 109.7, 98.5, 30.0, 29.7, 22.6, 13.9; MS (70 eV, EI) *m/z* (%): 276 (M$^+$, 1.33), 133 (100); IR (neat) *v*: 3030, 2957, 2930, 2869, 1930, 1697, 1600, 1454, 1383, 1169, 1013 cm^{-1}; HRMS calcd for *m/z* $C_{20}H_{20}O$ [M$^+$]: 276.1509; found 276.1513.

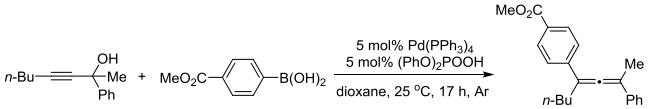
Preparation of 1-phenyl-3-(4-methoxylcarbonylphenyl)-1,2-heptadiene (3io, QAN-3-148).



1i **2o** (2.0 equiv) **3io**, 74%

Following **Typical Procedure**, the reaction of $Pd(PPh_3)_4$ (57.8 mg, 0.05 mmol), (4-(methoxycarbonyl)phenyl)boronic acid (360.1 mg, 2.0 mmol), $(PhO)_2POOH$ (12.5 mg, 0.05 mmol), **1i** (188.3 mg, 1.0 mmol), and anhydrous dioxane (5 mL) at 40 °C afforded **3io**[6h] (226.7 mg, 74%) as an oil [eluent: petroleum ether/ethyl acetate = 60/1 (240 mL) to petroleum ether/ethyl acetate = 30/1 (300 mL) to petroleum ether/ethyl acetate = 20/1 (200 mL)]; ^{1}H NMR (400 MHz, $CDCl_3$) δ: 7.97 (d, J = 8.4 Hz, 2H, Ar-H), 7.50 (d, J = 8.4 Hz, 2H, Ar-H), 7.35—7.27 (m, 4H, Ar-H), 7.24—7.18 (m, 1H, Ar-H), 6.58 (t, J = 2.8 Hz, 1H, =CH), 3.90 (s, 3H, OCH_3), 2.62—2.47 (m, 2H, CH_2), 1.64—1.53 (m, 2H, CH_2), 1.48—1.39 (m, 2H, CH_2), 0.92 (t, J = 7.2 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) δ: 207.5, 166.9, 141.2, 134.0, 129.8, 128.8, 128.5, 127.2, 126.8, 126.0, 109.6, 98.3, 52.0, 30.0. 29.7, 22.6, 13.9; MS (70 eV, EI) *m/z* (%): 306 (M$^+$, 1.5), 163 (100); IR (neat) *v*: 3029, 2954, 2930, 2867, 1931, 1719, 1604, 1435, 1409, 1185, 1016 cm^{-1}.

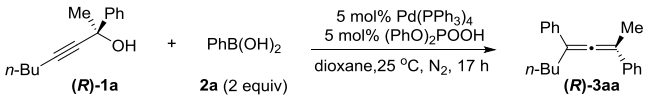
Gram scale synthesis of 2-phenyl-4-(4-methoxylcarbonylphenyl)-2,3-octadiene (3ao, QAN-3-119)



1a **2o** (2.0 equiv) **3ao**, 91%

To an oven-dried 100 mL Schlenk tube were added (4-(methoxycarbonyl)phenyl)boronic acid (1.7962 g, 10.0 mmol), $(PhO)_2POOH$ (62.6 mg, 0.25 mmol), and $Pd(PPh_3)_4$ (288.9 mg, 0.25 mmol). After replacing air with argon for three times at rt by vacuum, **1a** (1.0065 g, 5.0 mmol), and anhydrous dioxane (25 mL) were added sequentially. The resulting mixture was stirred for 17 h in oil bath at 25 °C and filtered through a short column of silica gel (3 cm) eluted with ethyl acetate (50 mL). After removal of the solvent under vacuum, the crude product was analyzed by ^{1}H NMR with CH_2Br_2 (70 μL) as the internal standard: 96% NMR yield of **3ao** was formed based on ^{1}H NMR analysis. The residue was purified by column chromatography on silica gel to afford **3ao** (1.4572 g, 91%) as a white solid [eluent: petroleum ether/DCM/Et_2O = 100/1/1 (~200 mL) to petroleum ether/DCM/Et_2O = 50/1/1 (~200 mL) to petroleum ether/DCM/Et_2O = 40/1/1 (420 mL)]; ^{1}H NMR (400 MHz, $CDCl_3$) δ: 7.96 (d, J = 8.4 Hz, 2H, Ar-H), 7.50—7.40 (m, 4H, Ar-H), 7.32 (t, J = 7.6 Hz, 2H, Ar-H), 7.25—7.18 (m, 1H, Ar-H), 3.89 (s, 3H, CH_3), 2.58—2.51 (m, 2H, CH_2), 2.22 (s, 3H, CH_3), 1.63—1.51 (m, 2H, CH_2), 1.48—1.35 (m, 2H, CH_2), 0.91 (t, J = 7.2 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) δ: 206.6, 167.0, 142.0, 136.6, 129.7, 128.4, 128.1, 126.9, 125.8, 125.7, 107.5, 104.2, 51.9, 30.0, 29.8, 22.5, 16.7, 13.9.

Preparation of (*R*)-2,4-diphenyl-2,3-octadiene ((*R*)-3aa, QAN-3-125)



(*R*)-**1a** 96% ee **2a** (2 equiv) (*R*)-**3aa** 55%, 2% ee

Following **Typical Procedure**, the reaction of $Pd(PPh_3)_4$ (11.6 mg, 0.01 mmol), phenylboronic acid (49.0 mg, 0.4 mmol),

 © 2020 SIOC, CAS, Shanghai, & WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim

16147065, 2020, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cjoc.201900442 by Zhejiang University, Wiley Online Library on [01/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

$(PhO)_2POOH$ (2.5 mg, 0.01 mmol), (*R*)-**1a** (40.3 mg, 0.2 mmol), and anhydrous dioxane (1 mL) at 25 °C afforded (*R*)-**3aa** (28.7 mg, 55%, 2% ee) as an oil [eluent: petroleum ether (200 mL)]; HPLC conditions: OJ-H column, hexane/*i*-PrOH = 200/1, 0.7 mL/min, λ = 214 nm, t_R (minor) = 6.2 min, t_R (major) = 7.5 min.

Biological activity study

GLP-1 measurement. Mouse intestinal cell line STC-1 was kindly provided by Dr. Jia Li (Shanghai Institute of Materia Medica, Chinese Academy of Sciences) and cultured in DMEM medium, containing 15% FBS, 100 U/mL penicillin and 100 μg/mL streptomycin at 37 °C in a humidified atmosphere of 5% CO_2. The cells were seeded in 96-well plates at a density of 40000 cells/well. After cultured overnight, the cells were treated with 0.1% DPP-IV inhibitor PK44 in KRB buffer containing 0.2% fatty acid-free BSA for 2 h. Cells were then treated with vehicle control (DMSO), positive control (1.0 μmol·L^{-1} PMA) or tested compounds for another 2 h. The supernatants of the culture medium were collected and the concentration of GLP-1 in the supernatants was measured by using active GLP-1 kit according to the manufacturer's instructions (Cisbio, Bedford, MA).

Glucose consumption. Rat L6 myotube cells were kindly provided by Dr. Jia Li (Shanghai Institute of Materia Medica, Chinese Academy of Sciences) and cultured in high-glucose (4.5 g/L) DMEM medium supplemented with 10% FBS, 100 U/mL penicillin and 100 μg/mL streptomycin. For differentiation, the cells were maintained in high-glucose (4.5 g/L) DMEM containing 2% FBS for 6 d and the medium was changed every 2 d. After differentiation, the myotube cells were incubated in serum-free, low-glucose (1.0 g/L) DMEM medium for 6 h and treated with vehicle control (DMSO), positive control (10 mmol·L^{-1} metformin) or tested compounds for 24 h. The concentration of glucose in medium was measured with glucose assay reagent (Mingdian, Shanghai, China) according to the glucose oxidase method.

Supporting Information

The supporting information for this article is available on the WWW under https://doi.org/10.1002/cjoc.201900442.

Acknowledgement

Financial support from National Natural Science Foundation of China (Grant No. 21690063 for S. Ma and Grant No. 21801041 for H. Qian), and Shanghai Sailing Program (18YF1402000 for H. Qian) are greatly appreciated. We thank Mr. Junjie Fan in this group for reproducing the results of **3ag**, **3aj** and **3ba** presented in this study.

References

[1] (a) *Modern Allene Chemistry*, Vol. 2, Eds.: Krause, N.; Hashimi, A. S. K., Wiley-VCH, Weinheim, Germany, **2004**; For selected reviews on transformations of allenes, see: (b) Zhang, X.; Ma, S. Fluorination of Allenes. In *Fluorination*, *Synthetic Organofluorine Chemistry 1*, Springer, Singapore, **2019**, pp. 1–16; (c) Zimmer, R.; Dinesh, C. U.; Nandanan, E.; Khan, F. A. Palladium-catalyzed reactions of allenes. *Chem. Rev.* **2000**, *100*, 3067–3126; (d) Ma, S. Transition Metal-Catalyzed/Mediated Reaction of Allenes with a Nucleophilic Functionality Connected to the α-Carbon Atom. *Acc. Chem. Res.* **2003**, *36*, 701–712; (e) Sydnes, L. K. Allenes from Cyclopropanes and Their Use in Organic Synthesiss Recent Developments. *Chem. Rev.* **2003**, *103*, 1133–1150; (f) Ma, S. Some Typical Advances in the Synthetic Applications of Allenes. *Chem. Rev.* **2005**, *105*, 2829–2872; (g) Hammond, G. B. Functionalized Fluorinated Allenes. *ACS Symp. Ser.* **2005**, *911*, 204–215; (h) Brasholz, M.; Reissig, H. U.; Zimmer, R. Sugars, Alkaloids, and Heteroaromatics: Exploring Heterocyclic Chemistry with Alkoxyallenes. *Acc. Chem. Res.* **2009**, *42*, 45–56; (i) Ma, S. Electrophilic Addition and Cyclization Reactions of Allenes. *Acc. Chem. Res.* **2009**, *42*, 1679–1688; (j) Yu, S.; Ma, S. Allenes in Catalytic Asymmetric Synthesis and Natural Product Syntheses. *Angew. Chem. Int. Ed.* **2012**, *51*, 3074–3112; (k) Lu, T.; Lu, Z.; Ma, Z. X.; Zhang, Y.; Hsung, R. P. Allenamides: A Powerful and Versatile Building Block in Organic Synthesis. *Chem. Rev.* **2013**, *113*, 4862–4904; (l) Ye, J.; Ma, S. Palladium-Catalyzed Cyclization Reactions of Allenes in the Presence of Unsaturated Carbon−Carbon Bonds. *Acc. Chem. Res*. **2014**, *47*, 989–1000; (m) López, F; Mascareñ, J. L. [4+2] and [4+3] catalytic cycloadditions of allenes. *Chem. Soc. Rev.* **2014**, *43*, 2904–2915; (n) Alcaide, B.; Almendros, P.; Aragoncillo, C. Cyclization reactions of bis(allenes) for the synthesis of polycarbo(hetero)cycles. *Chem. Soc. Rev.* **2014**, *43*, 3106–3135; (o) Muñoz, M. P. Silver and platinum-catalysed addition of O–H and N–H bonds to allenes. *Chem. Soc. Rev.* **2014**, *43*, 3164–3183; (p) Holmes, M.; Schwartz, L. A.; Krische, M. J. Intermolecular Metal-Catalyzed Reductive Coupling of Dienes, Allenes, and Enynes with Carbonyl Compounds and Imines. *Chem. Rev.* **2018**, *118*, 6026–6052; (q) Liu, Y.; Bandini, M. Nickel Catalyzed Functionalization of Allenes. *Chin. J. Chem.* **2019**, *37*, 431–441; (r) Liu, L.; Ward, R. M.; Schomaker, J. M. Mechanistic Aspects and Synthetic Applications of Radical Additions to Allenes. *Chem. Rev*. **2019**, *119*, 12422–12490.

[2] Hoffmann-Röder, A.; Krause, N. Synthesis and Properties of Allenic Natural Products and Pharmaceuticals. *Angew. Chem. Int. Ed.* **2004**, *43*, 1196–1216; *Angew. Chem*. **2004**, *116*, 1216–1236.

[3] For reviews on the synthesis of allenes, see: (a) Chapters 1–8 in ref. 1; (b) Neff, R. K.; Frantz, D. E. Recent Advances in the Catalytic Syntheses of Allenes: A Critical. *ACS Catal.* **2014**, *4*, 519–528; (c) Yu, S.; Ma, S. How Easy Are the Syntheses of Allenes? *Chem. Commun*. **2011**, *47*, 5384–5418; (d) Ogasawara, M. Catalytic Enantioselective Synthesis of Axially Chiral Allenes. *Tetrahedron*: *Asymmetry* **2009**, *20*, 259–271; (e) Brummond, K. M.; DeForrest, J. E. Synthesizing Allenes Today (1982–2006). *Synthesis* **2007**, 795–818; (f) Krause, N.; Hoffmann-Röder, A. Synthesis of Allenes with Organometallic Reagents. *Tetrahedron* **2004**, *60*, 11671–11694; (g) Huang, X.; Ma, S. Allenation of Terminal Alkynes with Aldehydes and Ketones. *Acc. Chem. Res*. **2019**, *52*, 1301−1312; (h) Fu, L.; Gressies, S.; Chen, P.; Liu, G. Recent Advances and Perspectives in Transition Metal-Catalyzed 1,4-Functionalizations of Unactivated 1,3-Enynes for the Synthesis of Allenes. *Chin. J. Chem*. **2020**, *38*, 91–100.

[4] For selected reviews on C-C bond formation via cross-coupling reactions, see: (a) Stille, J. K. The Palladium-Catalyzed Cross-Coupling Reactions of Organotin Reagents with Organic Electrophiles. *Angew. Chem. Int. Ed.* **1986**, *25*, 508–524; *Angew. Chem*. **1986**, *98*, 504–519; (b) Littke, A. F.; Fu, G. C. Palladium-Catalyzed Coupling Reactions of Aryl Chlorides. *Angew. Chem. Int. Ed.* **2002**, *41*, 4176–4211; *Angew. Chem*. **2002**, *114*, 4350–4386; (c) Yin, L.; Liebscher, J. Carbon-Carbon Coupling Reactions Catalyzed by Heterogeneous Palladium Catalysts. *Chem. Rev.* **2007**, *107*, 133–173; (d) Martin, R.; Buchwald, S. L. Palladium-Catalyzed Suzuki-Miyaura Cross-Coupling Reactions Employing Dialkylbiaryl Phosphine Ligands. *Acc. Chem. Res*. **2008**, *41*, 1461–1473; (e) Chen, X.; Engle, K. M.; Wang, D.-H.; Yu, J.-Q. Palladium(II)-Catalyzed C-H Activation / C-C Cross-Coupling Reactions: Versatility and Practicality. *Angew. Chem. Int. Ed*. **2009**, *48*, 5094–5115; *Angew. Chem*. **2009**, *121*, 5196–5217; (f) Yeung, C. S.; Dong, V. M. Catalytic Dehydrogenative Cross-Coupling: Forming Carbon-Carbon Bonds by Oxidizing Two Carbon-Hydrogen Bonds. *Chem. Rev*. **2011**, *111*, 1215–1292; (g) Milligan, J. A.; Phelan, J. P.; Badir, S. O.; Molander , G. A. Alkyl Carbon–Carbon Bond Formation by Nickel/Photoredox Cross-Coupling. *Angew. Chem. Int. Ed*. **2019**, *58*, 6152–6163; *Angew. Chem*. **2019**, *131*, 6212–6224; (h) Zheng, Y.; Miao, B.; Qin, A.; Xiao, J.; Liu, Q.; Li, G.; Zhang, L.; Zhang, F.; Guo, Y.; Ma, S. Negishi Coupling for Highly Selective Syntheses of Allenes via Ligand Effect and Mechanistic Study via SAESI-MS/MS. *Chin. J. Chem*. **2019**, *37*, 1003–1008; (i) Zheng, Y.; Miao, B.; Zhang, L. Intermediate Detection by SAESI-MS/MS Unveiling the Catalytic Cycle in Negishi Cou-

 © 2020 SIOC, CAS, Shanghai, & WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim

16147065, 2020, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cjoc.201900442 by Zhejiang University, Wiley Online Library on [01/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

pling. *Chin. J. Chem*. **2020**, *38*, 106–107.

[5] For selected reviews on Suzuki coupling reactions, see: (a) Miyaura, N.; Suzuki, A. Palladium-Catalyzed Cross-Coupling Reactions of Organoboron Compounds. *Chem. Rev*. **1995**, *95*, 2457–2483; (b) Polshettiwar , V.; Decottignies, A.; Len, C.; Fihri, A. Suzuki–Miyaura Cross-Coupling Reactions in Aqueous Media: Green and Sustainable Syntheses of Biaryls. *ChemSusChem* **2010**, *3*, 502–522; (c) Fihri, A.; Bouhrara, M.; Nekoueishahraki, B.; Basset, J.-M.; Polshettiwar, V. Nanocatalysts for Suzuki Cross-Coupling Reactions. *Chem. Soc. Rev*. **2011**, *40*, 5181–5203; (d) Han, F.-S. Transition-Metal-Catalyzed Suzuki–Miyaura Cross-Coupling Reactions: A Remarkable Advance from Palladium to Nickel Catalyst. *Chem. Soc. Rev*. **2013**, *42*, 5270–5298; (e) Hooshmand, S. E.; Heidari, B.; Sedghi, R.; Varma, R. S. Recent Advances in the Suzuki–Miyaura Cross-Coupling Reaction Using Efficient Catalysts in Eco-Friendly Media. *Green Chem*. **2019**, *21*, 381–405.

[6] (a) Moriya, T.; Miyaura, N.; Suzuki, A. Synthesis of Allenes by Palladium-Catalyzed Cross-Coupling Reaction of Organoboron Compounds with Propargylic Carbonates: Transmetallation of Organoboron Compounds with (Alkoxo)palladium Complexes Under Neutral Conditions. *Synlett* **1994**, 149–151; (b) Ishikura, M.; Agata, I. The Palladium Catalysed Cross-Coupling Reaction of Triethyl (1-Methylindol-2-yl)borate with Propargyl Carbonate. *Heterocycles* **1996**, *43*, 1591–1595; (c) Ishikura, M.; Matsuzaki, Y.; Agata, I.; Katagiri, N. Palladium Catalyzed Cross-Coupling Reaction of Indolylborates. *Tetrahedron* **1998**, *54*, 13929–13942; (d) Yoshida, M.; Ueda, H.; Ihara, M. Palladium-Catalyzed Coupling Reaction of Propargylic Oxiranes with Arylboronic Acids in Aqueous Media. *Tetrahedron Lett*. **2005**, *46*, 6705–6708; (e) Molander, G. A.; Sommers, E. M.; Baker, S. R. Palladium(0)-Catalyzed Synthesis of Chiral Ene-allenes Using Alkenyl Trifluoroborates. *J. Org. Chem*. **2006**, *71*, 1563–1568; (f) Yoshida, M.; Okada, T.; Shishido, K. Enantiospecific Synthesis of 1,3-Disubstituted Allenes by Palladium-Catalyzed Coupling of Propargylic Compounds with Arylboronic Acids. *Tetrahedron* **2007**, *63*, 6996–7002; (g) Luo, H.; Yu, Y.; Ma, S. Suzuki Coupling for Preparation of Allenes-Ligand Effects and Chirality Transfer. *Org. Chem. Front*. **2016**, *3*, 1705–1710; (h) Xiao, J.; Luo, H.; Huang, S.; Qian, H.; Ma, S. Tri(*o*-tolyl)phosphine for Highly Efficient Suzuki Coupling of Propargylic Carbonates with Boronic Acids. *Chem. Commun*. **2018**, *54*, 10451–10454; (i) Yao, Y.; Zhu, G.; Qin, C.; Qian, H.; Ma, S. Efficient Synthesis of Tetrasubstituted 2,3-allenoates and Preliminary Studies on Bioactivities. *Org. Chem. Front*. **2019**, *6*, 304–308; (j) Yoshida, M.; Gotou, T.; Ihara, M. Palladium-Catalyzed Direct Coupling Reaction of Propargylic Alcohols with Arylboronic Acids. *Tetrahedron Lett*. **2004**, *45*, 5573–5575; (k) Green, N. J.; Willis, A. C.; Sherburn, M. S. Direct Cross-Couplings of Propargylic Diol. *Angew. Chem. Int. Ed*. **2016**, *55*, 9244–9248; *Angew. Chem*. **2016**, *128*, 9390–9394.

[7] (a) Liu, N.; Zhi, Y.; Yao, J.; Xing, J.; Lu, T.; Dou, X. Rhodium(I)-Catalyzed Arylation/Dehydroxylation of tert-Propargylic Alcohols Leading to Tetrasubstituted Allenes. *Adv. Synth. Catal*. **2018**, *360*, 642–646. For Rh-catalyzed arylation of propargylic alcohols, see: (b) Wu, S.; Fu, C.; Ma, S. Asymmetric S_N2'-type C–H Functionalization of Arenes with Propargylic Alcohols. *Org. Chem. Front*. **2017**, *4*, 2002–2007.

[8] (a) Zhang, W.; Huang, C.; Yuan, Y.; Ma, S. Catalytic Transient Leaving Group for Atom-Economic Synthesis of Allenes from 2-alkynols. *Chem. Commun*. **2017**, *53*, 12430–12433; (b) Zheng, W.-F.; Zhang, W.; Huang, J.; Qian, H.; Ma, S. 2,3-Allenoic Acids via Palladium-Catalyzed Carboxylation of Propargylic Alcohols. *Org. Chem. Front*. **2018**, *5*, 1900–1904; (c) Zheng, W.-F.; Zhang, W.; Huang, C.; Wu, P.; Qian, H.; Wang, L.; Guo, Y.; Ma, S. Tetrasubstituted Allenes via the Palladium-Catalysed Kinetic Resolution of Propargylic Alcohols Using a Supporting Ligand. *Nat. Catal*. **2019**, *2*, 997–1005.

[9] Li, P.; Lü, B.; Fu, C.; Ma, S. Zheda-Phos for General α-Monoarylation of Acetone with Aryl Chlorides. *Adv. Synth. Catal*. **2013**, *355*, 1255–1259.

[10] Zhang, W.; Ma, S. Palladium/H^+ Cocatalyzed Kinetic Resolution of Tertiary Propargylic Alcohols. *Chem. Commun*. **2018**, *54*, 6064–6067.

[11] (a) Ogoshi, S.; Nishida, T.; Shinagawa, T.; Kurosawa, H. Key Process in Palladium-Catalyzed Asymmetric Transformation of Propargyl Electrophiles. Racemization of Optically Active η^1-Allenylpalladium(II). *J. Am. Chem. Soc.* **2001**, *123*, 7164–7165; (b) Wang, Y.; Zhang, W.; Ma, S. Identifying a Non-chiral Ligand for the Efficient Chirality Transfer in Carbonylation of Propargylic Mesylates. *Org. Chem. Front.* **2014**, *1*, 807–811.

[12] Pennell, M. N.; Foster, R. W.; Turner, P. G.; Hailes, H. C.; Tame, C. J.; Sheppard, T. D. *Chem. Commun*. **2014**, *50*, 1302–1304.

Manuscript received: November 6, 2019
Manuscript revised: January 2, 2020
Manuscript accepted: January 2, 2020
Accepted manuscript online: January 6, 2020
Version of record online: February 21, 2020

 © 2020 SIOC, CAS, Shanghai, & WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim

16147065, 2020, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cjoc.201900442 by Zhejiang University, Wiley Online Library on [01/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License